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Research Article

Locomotion control of a rigid-soft coupled snake robot in multiple environments

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ABSTRACT

The versatile motion capability of snake robots offers themselves robust adaptability in varieties of challenging environments where traditional robots may be incapacitated. This study reports a novel flexible snake robot featuring a rigid–flexible coupling structure and multiple motion gaits. To better understand the robot's behavior, a bending model for the soft actuator is established. Furthermore, a dynamic model is developed to map the relationship between the input air pressure and joint torque, which is the model base for controlling the robot effectively. Based on the wave motion generated by the joint coupling direction function in different planes, multiple motion gait planning methods of the snake-like robot are proposed. In order to evaluate the adaptability and maneuverability of the developed snake robot, extensive experiments were conducted in complex environments. The results demonstrate the robot's effectiveness in navigating through intricate settings, underscoring its potential for applications in various fields.

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1. Introduction

Currently, robots have widespread applications in diverse fields such as security, search and rescue, transportation, et al. [1,2]. Operating in narrow and confined spaces is one of the practical requirement for robots in these domains, necessitating specialized capabilities like compact size and strong environmental adaptability [3]. Among all types of robots, snake robots have emerged as a focal point of research due to their multi-gait movement ability and adaptability to complex environments. Thus, researchers have long been engaged in the study of robots that can emulate snake-like motions.

In 1972, Hirose developed the first serpentine robot ACM-III that could move steadily over rough surfaces or winding paths due to its slender and flexible body [4]. Subsequently, Hirose conducted experimental studies on biological snakes, including detailed quantitative analyses of typical snake movement patterns [5]. Based on Hirose's groundbreaking work in both biological research methods and the development of snake robot prototypes [6–8], extensive theoretical and experimental investigations into snake robots have been conducted [9–11].

Biological snakes employ a variety of gaits to adeptly navigate ever-changing environmental conditions, inspiring the necessity for a diverse range of locomotion gaits in robotic snakes, a pivotal focus in snake robot research [12]. The most common biological snake gaits are lateral undulation gait, sidewinding gait, lateral roll gait and linear gait [13]. Biological snakes can autonomously employ the most suitable gait according to their surroundings and specific state. In different confined spaces, snakes can adopt various forms, such as approaching targets through linear movements during hunting, or employing sidewinding gait on loose sand [14]. Various serpentine robots, such as ACM-III [4], AmphiBot I/II [15] and Aiko [16], utilized a lateral wave gait for movement. While Modsnake [17] and Kulko [18] robots imitated the side-moving state of biological snakes. LocoSnake [19] and Graphene-Based snake robot [20] used accordion gait to explore narrow and restricted pipelines. Nevertheless, Bayraktaroglu's snake robot [21] realized efficient movement through linear gait. Various limb-free gaits offer valuable insights for the study of snake robot locomotion [5]. For example, modular climbing caterpillar robot [22], MMIR-V1/V2 [23] imitates the inchworm movement gait. And Chen designed a snake robot with Horizontal rolling gait [24]. Moreover, Hoshino proposed a snake robot to achieve jumping gait, thereby establishing the foundation for implementing jumping gaits in snake robots [25]. The snake robot Modsnake can realize three locomotion gaits: lateral movement, climbing and underwater swimming, which enables it to adapt to varied and rugged terrain [17]. And the lateral movement and lateral rolling gaits of Kulko enables itself to actively avoid obstacles [18]. Slim slime robot (SSR) can also realize three locomotion gaits: lateral wave, straight line and lateral roll, which enables it to traverse

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complex terrain and operate in narrow Spaces [26]. It is obvious that Snake robots with multiple motion gaits often exhibit superior adaptability and maneuverability compared to those with a single motion gait, especially in complex environments. Thus, understanding the principle of various motion gaits in snake robots can certainly enhance their adaptability and maneuverability in complex environments.

A comprehensive investigation [13,15,27], including theoretical modeling, structural design, and control aspects of rigid snake robots, have shown that rigid snake robots can achieve efficient and stable motion across various terrains [17]. However, when compared to biological snakes, they still exhibit limitations in environmental adaptability. Biological snakes, with their soft bodies, demonstrate exceptional terrain navigation capabilities. The inherent softness enables them to traverse through narrow openings and adapt without causing harm to the environment or themselves [28]. With the advancement of flexible material technology, there is an opportunity to leverage the benefits of flexible materials to enhance the movement and environmental adaptability of snake robots. The pliability of flexible materials allows for the generation of more lifelike movements during the driving process, thereby significantly improving the environmental adaptability of snake robots. Additionally, the compact structure of flexible materials liberates snake robots from some intricate rigid structures, providing distinct advantages in navigating through restricted environments [29,30].

At present, there are several soft snake robots developed using flexible materials, including a compact snake robot with a length of only 0.185 m [31], a snake robot capable of maneuvering through narrow tubes [32], a versatile snake robot designed for various environments [33] and an underwater-swimming snake robot [34]. The operational environments and modes of movement of these soft snake robots closely resemble those of biological snakes, and some demonstrate adaptability to different surroundings. However, these soft snake robots are often tailored to specific scenarios, excelling in particular forms of movement within those defined contexts. When confronted with changing requirements or environments, there is a possibility that they may fail to produce effective movement [35]. While certain robotic snakes can perform movements in various environmental scenarios, it is essential to emphasize that this does not necessarily indicate genuine environmental adaptability in these soft snake robots.

In order to enhance environmental adaptability, a snake robot named "ZJUT Snake" is designed with a rigid–flexible coupling structure in this paper, enabling the execution of multiple motion gaits. The contributions of this work are summarized as follows:

(1) New mechanism and structure design of a soft-actuator-driven snake robot: The soft actuator enables flexible and stable movement, allowing the ZJUT Snake to achieve three-dimensional motion.

(2) Gait analysis method for snake robot Movement: In response to the diverse forms of snake robot movement, a gait analysis method is introduced. This method subdivides the movement gait into visual steps, analyzes the impact of each step on overall movement, and summarizes the rules governing the ZJUT Snake's attainment of various movement gaits. This lays the foundation for gait control.

(3) Realization of three locomotion gaits in the soft snake robot: Three distinct locomotion gaits were successfully implemented in the ZJUT Snake, validating its adaptability to environments with different gaits. This achievement enhances the ZJUT Snake's capability to navigate complex environments by employing multiple gaits.

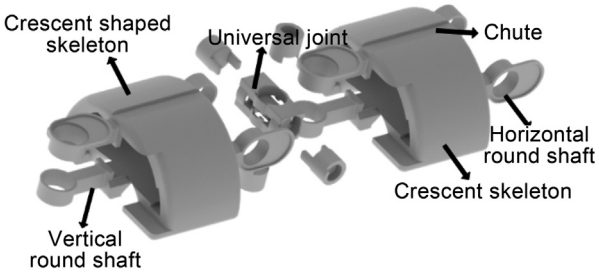


Fig. 1. Exploded view of rigid skeleton.

2. Structural design

The snake robot designed in this paper incorporates a novel concept of rigid-soft coupling, employing a modular design that integrates a rigid skeleton as its foundational structural unit and soft actuators as the propulsive driving component. The robot's dynamic motion is orchestrated by these soft actuators, synergizing the merits of both rigid and soft materials. This innovative combination enhances the structural strength and stability of the soft actuators, allowing the snake robot to maintain the flexibility and safety associated with soft robotics while also benefiting from the structural strength typically found in rigid robots.

2.1. Rigid skeleton structure

A rigid skeleton structure is proposed that can be installed inside a soft actuator, providing internal support to enhance the structural strength of the soft actuator. During the control of the soft actuator's motion, the inherent high flexibility and extensive degrees of freedom of the rigid skeleton ensure that it does not adversely affect the motion of the soft actuator. Additionally, by amounting sensors on the rigid skeleton, its motion state variables could be acquired. This approach facilitates the collection of motion data for soft materials, which is often challenging in soft robotics.

As shown in Fig. 1, the rigid skeleton structure is installed within the ten-chamber driving structure of the soft actuator. The overall rigid skeleton structure is composed of multiple crescent-shaped shells interconnected by universal joints. Vertical and horizontal round shafts are designed to facilitate the connection between crescent-shaped skeletal structures.

2.2. Soft actuator

The soft actuator, depicted in Fig. 2(a), comprises three primary components. The first part serves as a connection point and functions as the inflation inlet. External air pressure is introduced into the actuator through this inflation inlet, enabling control over the bending or elongation of the soft actuator in various directions. The second part consists of a ten-chamber driving unit, composed of ten pneumatic artificial muscles arranged in a specific geometric pattern, as illustrated in Fig. 2(b). The fundamental unit of the ten-chamber driving structure, the pneumatic artificial muscle, operates based on air pressure input. It is designed with an internal silicone core covered by an elastic polyester fabric. When air pressure is applied, the internal silicone expands and elongates, and constrained by the external polyester material, the pneumatic artificial muscle either elongates axially or expands radially. For this study, axial elongation-type pneumatic artificial muscles were utilized, extending axially with air pressure input.

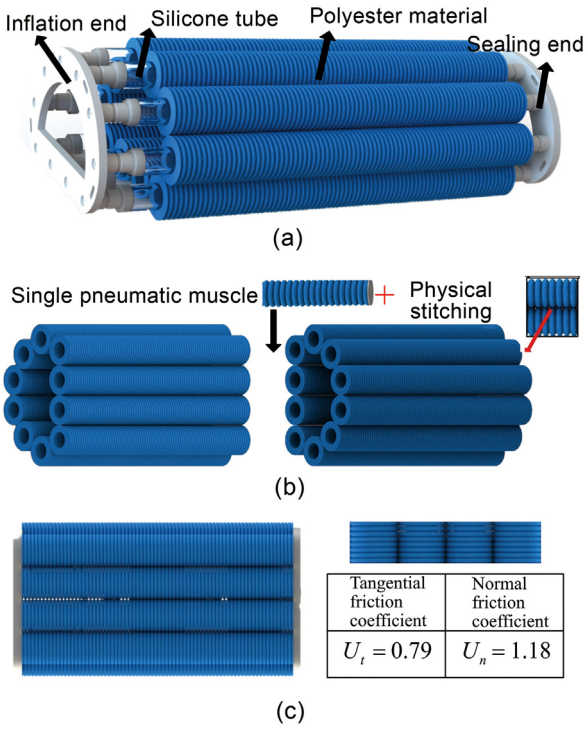


Fig. 2. Structure diagram of a snake robot. (a) Ten-cavity drive structure. (b) Ten cavities made of polyester. (c) Bottom of ten-cavity drive structure.

Upon inflating the ten-chamber driving structure, the coordinated action of the pneumatic artificial muscles induces a collective bending effect on the overall structure. The final component of the soft actuator serves as a connection point and fulfills a sealing function. This distinctive structural design of the soft actuator embodies three pivotal features: (1) Inflating the pneumatic artificial muscles empowers the soft actuator module to achieve motion in both horizontal and vertical directions, enabling the snake robot to execute three-dimensional gaits. (2) The bottom of the soft actuator introduces anisotropic friction. As the snake robot moves on the ground, it generates greater normal friction than tangential friction. This design allows the snake robot to traverse the ground, a critical element for generating forward propulsion in this soft snake robot. (3) The structural design of the soft actuator ensures exceptional mechanical stability. The flat bottom and elliptical body shape prevent the snake robot from tipping over during three-dimensional gaits, ensuring stability throughout its motion, even when swimming in water.

2.3. Motion principle of soft actuator

The mechanism of soft actuator elongation and bending induced by aerated artificial muscles is elucidated. Furthermore, we explore how precise control of air pressure input to the pneumatic artificial muscles enhances the soft actuator's capability to execute bends in both the horizontal and vertical planes.

The geometric configuration of the soft actuator is illustrated in Fig. 3(a), with A_1 to A_{10} representing ten pneumatic artificial muscles. Pneumatic artificial muscles A_1 to A_6 are arranged in a standard circular arc formation, with their centers oriented towards the geometric center O. Simultaneously, pneumatic artificial muscles A_7 to A_{10} are linearly distributed, with A_7 and A_{10} positioned externally and tangent to A_6 and A_1, respectively. The geometric centers of pneumatic artificial muscles A_7 and A_{10} are designated as points C and B, respectively.

Taking pneumatic artificial muscle A_3 as an example, we analyze the impact of a single pneumatic artificial muscle on the soft actuator after inflation. As depicted in Fig. 3(b), when air pressure is introduced into A_3, it experiences axial elongation. Structurally, A_2, A_3, and A_4 are mutually fixed tangentially. Consequently, when A_3 elongates axially, it stretches A_2 and A_4, causing them to move closer to one side of A_3. This stretching generates a tension force exerted by A_3 on A_2 and A_4, with the force directed towards A_3O.

Simultaneously inflating multiple pneumatic artificial muscles enables the attainment of bending in both the horizontal and vertical planes of the soft actuator. As illustrated in Fig. 3(c), when pneumatic artificial muscles A_1, A_2, and A_{10} are inflated concurrently, the directions of the forces acting on each pneumatic artificial muscle are as follows:

$$\overrightarrow{F_{A2}} = \cos\frac{\pi}{4}\cdot\vec{x} - \sin\frac{\pi}{4}\cdot\vec{y} \tag{1}$$

$$\overrightarrow{F_{A1}} = \cos\frac{\pi}{12}\cdot\vec{x} - \sin\frac{\pi}{12}\cdot\vec{y} \tag{2}$$

$$\overrightarrow{F_{A10}} = \cos\frac{11}{36}\pi\cdot\vec{x} + \sin\frac{11}{36}\pi\cdot\vec{y} \tag{3}$$

The direction of resultant F_1 is expressed as

$$\overrightarrow{F_1} = \overrightarrow{F_{A1}} + \overrightarrow{F_{A2}} + \overrightarrow{F_{A10}} = \left(\cos\frac{\pi}{12} + \cos\frac{\pi}{4} + \cos\frac{11\pi}{36}\right)\cdot\vec{x} + \left(\sin\frac{11\pi}{36} - \sin\frac{\pi}{12} - \sin\frac{\pi}{4}\right)\cdot\vec{y} \tag{4}$$

The inverse tangent function of $\overrightarrow{F_1}$ gives as $\arctan\overrightarrow{F_1} = -3.7°$. This also indicates that at this point, the angle between the direction of bending of the soft actuator and the X-axis is approximately $-3.7°$. To simplify the model of the soft actuator, we consider its motion under this inflation configuration as bending to the right. Due to the left–right symmetry in the structure of the soft actuator, when pneumatic artificial muscles A_5, A_6, and A_7 are simultaneously inflated, the direction of motion for the actuator is to the left. As shown in Fig. 3(d), when air pressure is injected into the pneumatic artificial muscles A_3 and A_4 at the same time:

$$\overrightarrow{F_{A3}} = \cos\frac{5}{12}\pi\cdot\vec{x} - \sin\frac{5}{12}\pi\cdot\vec{y} \tag{5}$$

$$\overrightarrow{F_{A4}} = \cos\frac{7}{12}\pi\cdot\vec{x} - \sin\frac{7}{12}\pi\cdot\vec{y} \tag{6}$$

The direction of the resultant F_2 is obtained by adding the two together:

$$\overrightarrow{F_2} = \overrightarrow{F_{A3}} + \overrightarrow{F_{A4}} = -2\sin\frac{5}{12}\pi\cdot\vec{y} \tag{7}$$

$\overrightarrow{F_2}$ has no X-direction component and is a negative value, which indicates that the soft actuator is bending in the opposite direction of the Y-axis, that is, bending down. As depicted in Fig. 2(e), when pneumatic artificial muscle A_8 and A_9 are charged with air pressure at the same time, the direction of $\overrightarrow{F_8}$ and $\overrightarrow{F_9}$ is pointing to the positive direction of the Y-axis, and the direction of the resultant force is also pointing to the Y-axis square, which indicates that the soft actuator is bending in the positive direction of the Y-axis at this time, that is, bending upward. Through these four inflation configurations involving multiple pneumatic artificial muscles, the soft actuator can achieve bending in both horizontal and vertical planes. If air pressure is simultaneously applied to the soft actuator to induce horizontal and vertical bending, the actuator will produce a combined effect of bending in both directions. When air pressure is simultaneously applied to all pneumatic artificial muscles, the soft actuator undergoes axial elongation. The bending effects of the soft actuator model are depicted in Fig. 4.

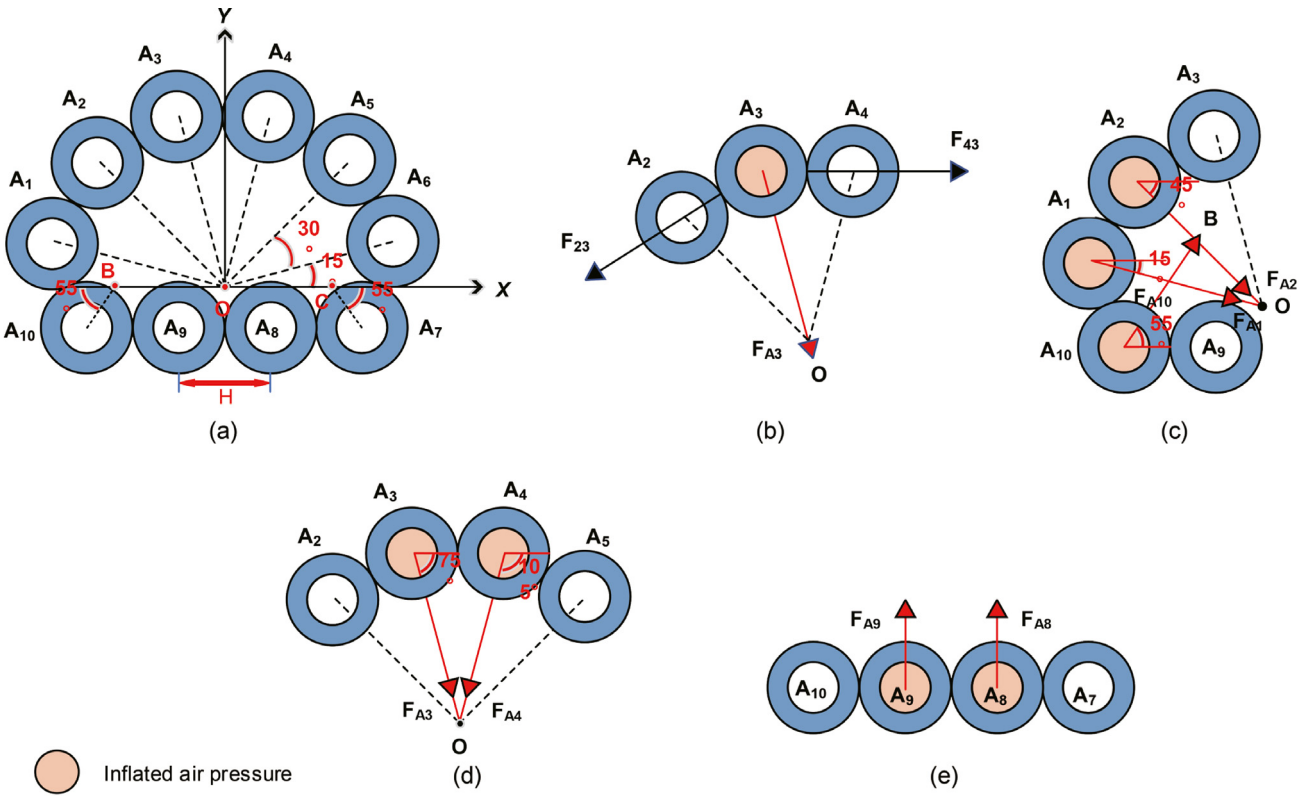


Fig. 3. Force analysis of soft actuator under different inflating modes.

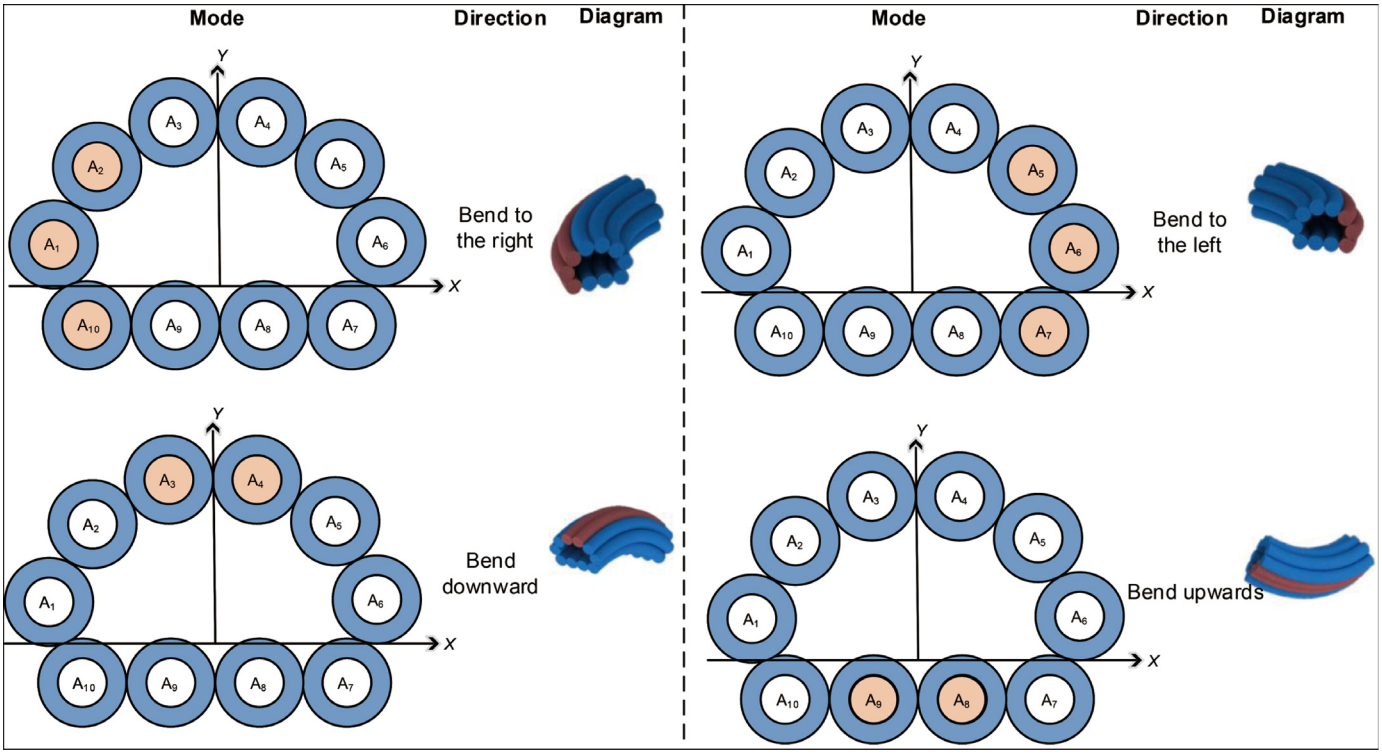


Fig. 4. The bending diagram of the soft actuator in four directions.

3. Locomotion control

By employing joint coupling direction functions to control the interconnected soft actuators on the snake robot, we achieved five distinct motion gaits: lateral undulation, sidewinding translation, lateral rolling, longitudinal traveling wave, and straight-line extension. Among these gaits, lateral undulation and longitudinal traveling waves are two-dimensional gaits occurring within one

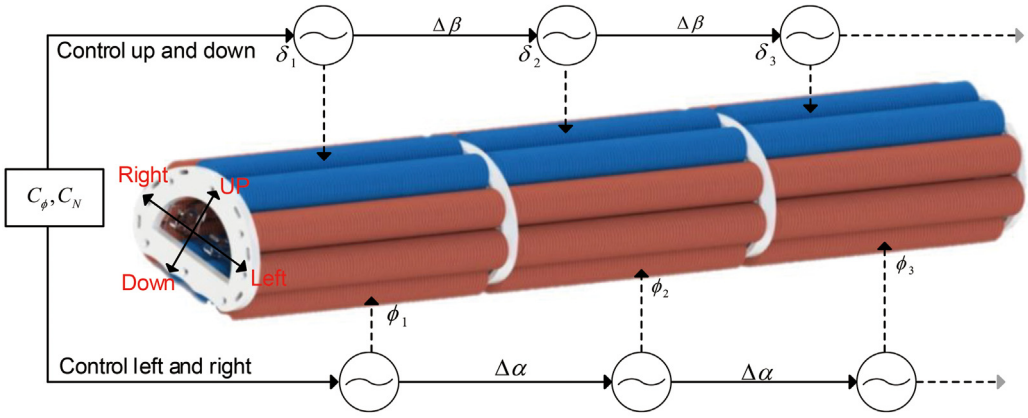


Fig. 5. Diagram of joint coupling function.

plane. Sidewinding translation, lateral rolling, and straight-line extension represent three-dimensional gaits, involving motion in two planes simultaneously. The diversity in the forms of these three-dimensional gaits liberates the snake robot's motion from environmental constraints, allowing it to adapt its gaits during movement according to the changing environment.

3.1. Joint coupling direction function

The control function in this paper is designed based on the Serpenoid control function for snake robot [36]. The control signals are generated in two orthogonal planes, as depicted in Fig. 5. The soft actuators serve as the driving joints of the snake robot, expressing them as functions of amplitude, period, offset, and phase in parameter space.

$$\begin{aligned}\phi_i &= D_\phi + A_\phi \sin(\frac{2\pi}{T_\phi}t + \alpha_0 - (i-1)\Delta\alpha)\\ \delta_i &= D_\delta + A_\delta \sin(\frac{2\pi}{T_\delta}t + \beta_0 - (i-1)\Delta\beta)\end{aligned} \tag{8}$$

where, joint yaw Angle ϕ_i is the left and right deflection angle of the joint driven by segment i in the horizontal direction, and joint pitch angle δ_i is the up and down deflection angle of the joint driven by segment i in the vertical direction. D, A and T in the joint coupling direction function represent offset, amplitude and period, respectively. α_0 and $\Delta\alpha$ represent the initial phase and phase lag of the drive joint in the direction of horizontal swing. β_0 and $\Delta\beta$ represent the initial phase and phase lag of the drive joint in the direction of vertical swing. The subscript i represents the joint position numbered from the head joint of the snake robot.

The positive or negative of the joint coupling direction function does not represent the size but the bending direction of the driving joint. When $\phi_i > 0$, the driving joint bends to the right in the horizontal direction; When $\phi_i < 0$, the driving joint bends to the left in the horizontal direction; When $\delta_i > 0$, the driving joint bends downward in the vertical direction; When $\delta_i < 0$, the driving joint bends upward in a vertical direction. Through the joint coupling direction function, we can control the snake robot to move in a specific manner in both the horizontal and vertical directions. The movement in these two directions is crucial for the snake robot's progression, as the interaction with the ground generates frictional forces that aid in its movement. The snake robot exhibits various gaits, and the roles of different parameters of the joint coupling direction function are described below.

Amplitude A: represents the maximum deflection angle of each drive joint in the horizontal or vertical direction, which varies from 0 to $\pi/2$. In a joint coupled direction function, it represents the degree of fluctuation (vertical or horizontal) of a segmented sine function.

Period T: determines the execution speed of the gait. During a gait cycle, all shapes (gestures) of any gait are performed in a continuous manner, after which the process is repeated.

Joint shift D: It will affect the direction of movement of the snake robot. When the average value of joint offset is 0, the snake robot will move in a straight line. When the average value of the joint offset is greater than 0, the snake robot will deflect counter-clockwise along the direction of motion. When the average value of the joint offset is less than 0, the snake robot will be deflected clockwise along the direction of motion.

Initial phase and phase lag: The initial phase affects the initial state of the sine wave, and the phase lag determines how the sine wave propagates along the snake robot, which can be stationary or propagated forward or backward.

3.2. Sidewinding state

This lateral motion gait is commonly employed by snakes residing in deserts or when encountering specific road conditions such as obstacles or unfavorable terrain. In this gait, the snake's interaction with the ground, achieved by swaying its body, aims not to move forward but rather to traverse sideways. To achieve the lateral motion gait, it is essential to control the snake robot to oscillate horizontally with a systematically changing yaw angle in the horizontal plane. Simultaneously, the pitch angle needs to be controlled to lift the snake robot's body off the ground periodically and adhere to the ground consistently in the vertical direction. When a specific joint of the snake robot moves horizontally in one direction, the joint is raised off the ground, generating a lateral offset. Conversely, when a particular joint moves horizontally in the opposite direction, it adheres to the ground, increasing friction, thus minimizing or eliminating the lateral offset. The joint coupling direction function under the lateral motion gait is expressed as

$$\phi_i = \frac{1}{3}\pi \sin(\frac{2\pi}{12}t + (i-1)\frac{1}{3}\pi) \tag{9}$$

$$\delta_i = \frac{1}{9}\pi \sin(\frac{2\pi}{12}t + \frac{\pi}{2} + (i-1)\frac{2}{3}\pi) \tag{10}$$

Utilizing joint coupling direction functions, the gait diagram for the snake robot's sidewinding locomotion is depicted in Fig. 6. The sidewinding gait is segmented into six steps (excluding the initial state and the subsequent gait cycle), where steps 1 to 6 collectively form one gait cycle. In the sidewinding locomotion gait, the snake robot, with its tail joint or head joint making

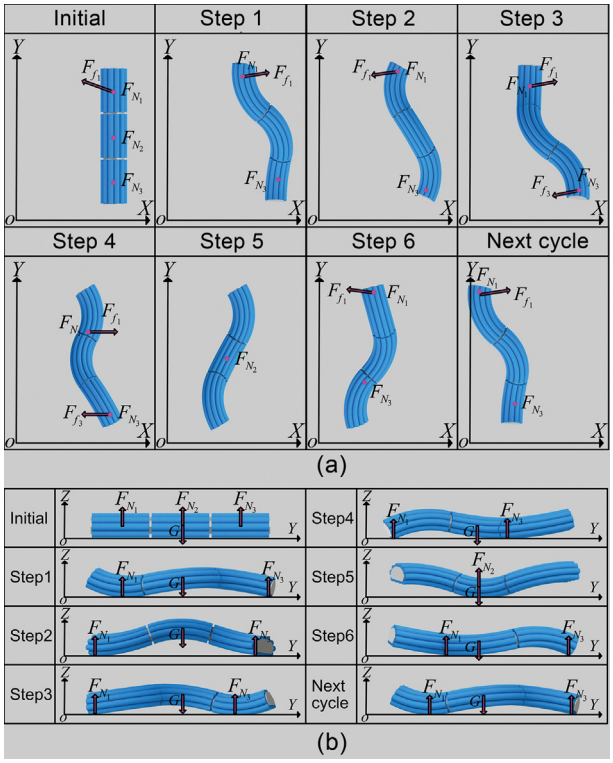


Fig. 6. The steps of sidewinding gait. (a) Top view of lateral roll gait. (b) Right view of lateral roll gait.

contact and adhering to the ground, utilizes the middle joint to propel the head joint in a specific direction. This mechanism enables the snake robot to maintain fixation of its tail segment while facilitating movement of the middle and head segments. Subsequently, the snake robot adjusts its posture to achieve fixation of the head segment while permitting movement of the middle and tail segments.

From the initial state to step 2: The snake robot's tail joint bends upward to adhere to the ground, maintaining a fixed position with slight deviations. The middle joint bends downward to lift off the ground, while driving the head joint to the left, enabling the snake robot to move in that direction.

From step 2 to step 3: This is a transitional pose change. The snake robot adjusts its head joint by bending it upward to adhere to the ground, and the middle and tail joints bend downward, preparing for the subsequent tail movement.

From step 3 to step 5: The snake robot's head joint remains fixed with slight deviations. The middle joint drives the tail joint to bend to the left, causing this portion of the snake robot to move leftward.

From step 5 to step 6: This is another transitional pose change. The snake robot adjusts its posture in preparation for head segment movement.

At this point, the snake robot completes a full cycle of sidewinding locomotion involving head movement, tail movement, and two transitional posture adjustments. By continuously repeating steps 1 to 6, the snake robot can achieve uninterrupted sidewinding movement. In the side-moving state, the robot and the ground also produce various opposite friction forces:

$$\begin{pmatrix} F_{fi,x} \\ F_{fi,y} \end{pmatrix} = - \begin{pmatrix} U_t(\cos\phi_i)^2 + U_n(\sin\phi_i)^2 & (U_t - U_n)\cos\phi_i \sin\phi_i \\ (U_t - U_n)\cos\phi_i \sin\phi_i & U_t(\sin\phi_i)^2 + U_n(\cos\phi_i)^2 \end{pmatrix} \begin{pmatrix} \dot{X}_{fi} \\ \dot{Y}_{fi} \end{pmatrix} \tag{11}$$

where U_t is the friction coefficient in the tangential direction, U_n is the friction coefficient in the normal direction, X_f and Y_f represent the velocity at the center coordinate of the ith joint.

$$F_{f1,x} + F_{f2,x} + F_{f3,x} = M\ddot{X}_G \tag{12}$$

$$F_{f1,y} + F_{f2,y} + F_{f3,y} = M\ddot{Y}_G \tag{13}$$

where X_G and Y_G are the acceleration of the robot's centroid coordinates.

3.3. Rolling gait

The rolling motion is a composite gait that amalgamates both horizontal and vertical movements of the snake robot, enabling it to maneuver flexibly in two orthogonal planes: horizontal and vertical. In this gait, the snake robot's roll and pitch angles undergo periodic sinusoidal oscillations. The snake robot consistently bends in a specific direction, leading to interaction forces between the driving joints and the ground. These interaction forces generate torque, initiating the rolling motion. Through the lateral rolling gait, the snake robot can execute sideward rolling and surmount certain ground obstacles. In the lateral rolling gait, the joint coupling direction functions are outlined as follows:

$$\phi_i = \frac{1}{6}\pi \sin\frac{2\pi}{12}t \tag{14}$$

$$\delta_i = \frac{1}{4}\pi \sin(\frac{2\pi}{12}t + \frac{1}{2}\pi) \tag{15}$$

Through joint coupling direction functions, the stages of the rolling gait are depicted in Fig. 7. The rolling gaits consist of eight steps (excluding the initial state and the subsequent cycle gait), with steps 1 to 8 constituting one cycle of lateral rolling gait. In the lateral rolling gait, the snake robot bends itself in one direction, enhancing ground support by flexing the driving joints toward the ground. Consequently, this motion induces the snake robot to perform a roll. Here is a step-by-step breakdown.

From the initial state to step 1: All driving joints of the snake robot bend to the left. The direction of the frictional force opposes the movement direction of the driving joints.

From step 1 to step 2: All driving joints bend downward. As the snake robot is in a left-bent position, the ground support forces F_{N1}, F_{N2}, and F_{N3} do not align on the same line. The robot experiences a clockwise torque. When the middle joint briefly lifts off the ground, the robot is only supported by F_{N1} and F_{N3}, and the torque reaches its maximum. When the torque is significant enough, the robot flips clockwise. After a certain angle, the middle joint recontacts the ground, and the head and tail joints lift off, achieving a momentary balance.

From step 2 to step 3: The snake robot gradually returns from the left-bent position to a straight line but increases its downward bending angle. In the Step 3 view of Fig. 7, the snake robot increases its leftward bending amplitude and the head and tail joints no longer lift off, returning to contact with the ground and preparing for the next flip action.

From step 3 to step 4: The snake robot bends to the right. In the Step 3 view of Fig. 7, the robot is bending toward the ground, and this bending generates a clockwise torque, driving the robot to flip clockwise.

From step 4 to step 5: The snake robot gradually returns from the downward bent position to a straight line and begins bending to the right. In the step 5 view of Fig. 7, the robot resumes ground contact and bends to the left, preparing for the next flip action.

From step 5 to step 6: The snake robot bends upward. In the step 5 view of Fig. 7 the robot bends toward the ground, and this bending generates a torque that drives the robot to flip clockwise.

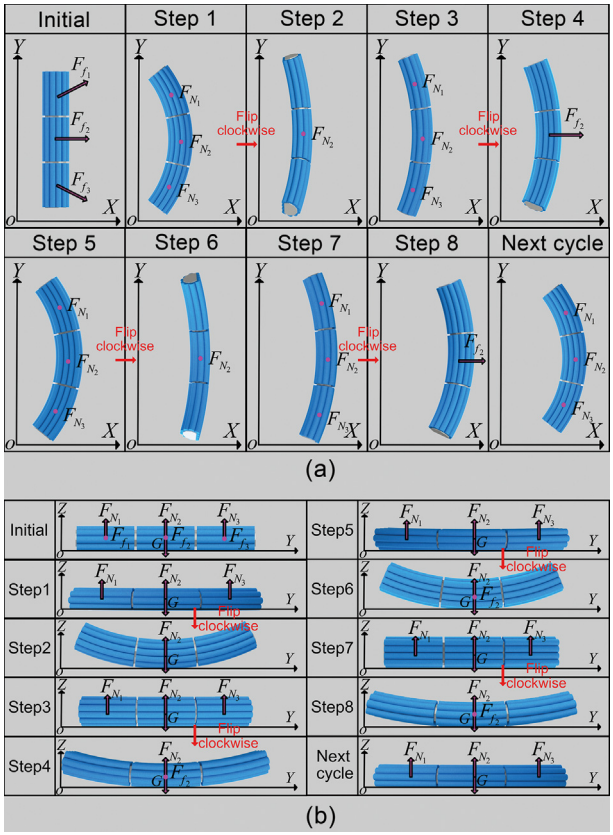


Fig. 7. Steps of rolling gait.

From step 6 to step 7: The snake robot gradually returns from the right-bent position to a straight line and increases its upward bending angle. In the step 7 view of Fig. 7, the robot resumes ground contact and increases its leftward bending angle, preparing for the next flip action.

Finally, from step 7 to step 8: The snake robot bends to the left. In the step 7 view of Fig. 7, the robot bends toward the ground, generating a torque that drives the robot to flip clockwise. At this point, the robot completes one full cycle of lateral rolling gait, having performed a full clockwise rotation. In the next cycle, the robot repeats these steps continuously to achieve the lateral rolling gait.

In the rolling gait, various external forces act on the robot. These include gravity (G), ground support force (F_N), and ground friction force (F_f). During several steps of the robot's flipping motion, the robot experiences a balance between the support force and gravity. Due to the symmetrical relationship between the support forces F_{N1} and F_{N3}, at this point.

$$F_{N_1} = F_{N_3} = \frac{1}{4}G \tag{16}$$

$$F_{N_2} = \frac{1}{2}G \tag{17}$$

The torque M of the robot is

$$M = \frac{1}{2}(X_{N_2} - X_{N_1})G \tag{18}$$

When the middle joint of the robot lifts off the ground, it is solely acted upon by the support forces F_{N1} and F_{N3}. At this moment, the robot experiences the maximum torque, denoted as M_{max}.

$$F_{N_1}{}' = F_{N_3}{}' = \frac{1}{2}G \tag{19}$$

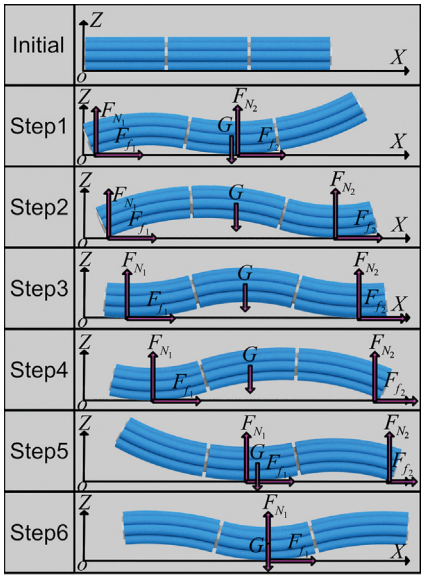


Fig. 8. Right view of longitudinal traveling wave gait steps.

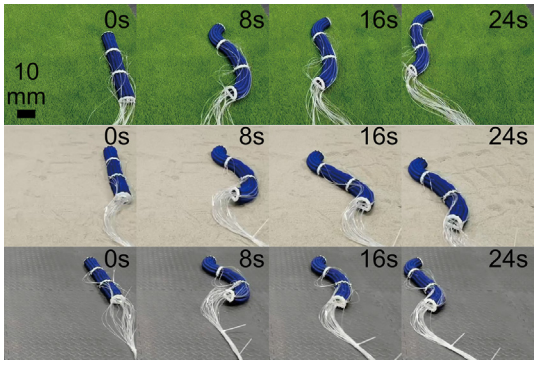


Fig. 9. Sidewinding-step in three scenarios.

$$M_{\max} = (X_G - X_{N_1})G \tag{20}$$

3.4. Longitudinal traveling wave gait

The longitudinal traveling wave gait is another locomotion pattern employed by snake robots. In terms of motion, it adapts the lateral wave motion gait to the vertical plane, requiring less space for movement, making it well-suited for narrow and constrained environments. In this gait, it is essential to control the pitch angle of the snake robot to generate a forward-propagating sine wave, enabling the robot to execute regular swings in the vertical plane. The joint coupling direction function for the longitudinal traveling wave gait is as follows:

$$\delta_i = \frac{1}{2}\pi + \frac{2}{3}\pi \sin\left(\frac{2\pi}{12}t + (i-1)\frac{2}{3}\pi\right) \tag{21}$$

Through the joint coupling direction function, the steps of longitudinal traveling wave gait are shown in Fig. 8. The longitudinal traveling wave gait has 6 steps, excluding the robot's initial state. During the longitudinal traveling wave gait, the snake robot transforms its body shape into a continuously propagating sine wave in the vertical plane. It achieves this by generating rolling friction with the ground at the bottom of each joint, propelling itself forward.

From the initial state to step 1, the snake robot raises its head joint and uses the friction between the middle joint and

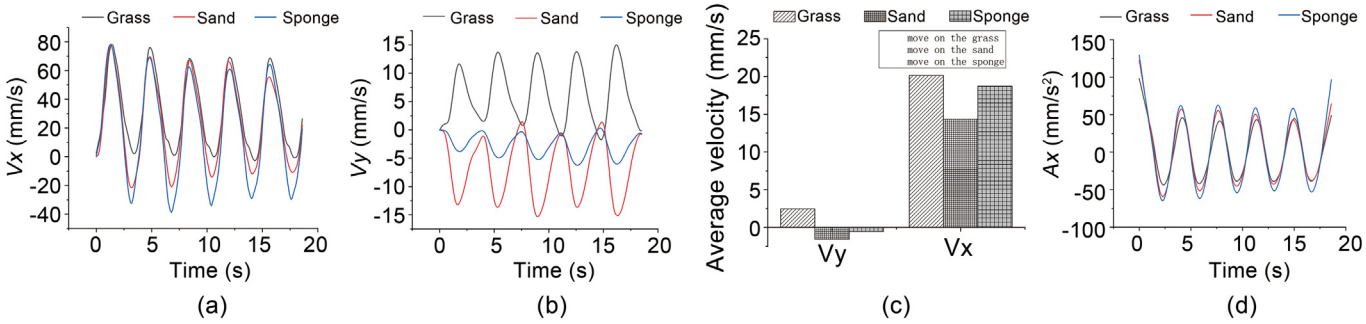


Fig. 10. Motion state of sidewinding.

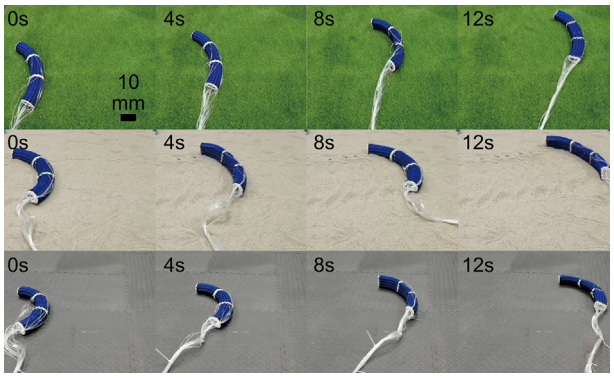


Fig. 11. Rolling gait in three scenarios.

the ground to generate forward propulsion. The tail joint only supports the ground, providing balance.

From step 1 to step 2, the snake robot lifts the middle joint and uses the friction between the head joint and the ground for propulsion. At this stage, the tail joint still serves to maintain balance.

From step 2 to step 3, the robot simultaneously contacts the ground with both the tail joint and the head joint, creating friction for self-propulsion.

From step 3 to step 4, the robot generates propulsion through the friction between the tail joint and the ground, while the head joint supports the ground to maintain balance.

From step 4 to step 5, the robot uses the friction between the middle joint and the ground for propulsion.

From step 5 to step 6, the robot adjusts its posture, keeping the middle joint in contact with the ground and lifting both the head and tail joints off the ground, smoothly transitioning to the next gait cycle.

The external forces applied to the robot under the longitudinal traveling wave gait are gravity G, ground support force F_N and ground friction force F_f, respectively. Gravity and support force are balanced, and the combined force of the robot is friction force. Therefore, friction force is the driving force of the robot, and the balance of force and torque can be obtained:

$$F_{N_1} = (G + M\ddot{Z}_G)\frac{X_{N_2} - X_G}{X_{N_2} - X_{N_1}} + M\ddot{X}_G\frac{Z_G}{X_{N_2} - X_{N_1}} \tag{22}$$

$$F_{N_2} = G - F_{N_1} \tag{23}$$

where M is the mass of the robot, the position of the centroid coordinates of the robots X_G and Z_G under the XOZ plane, MX_G and MZ_G are the inertial forces of the robot along the X and Z axes, X_{N1} and X_{N2} are the X-axis coordinates of the supporting forces F_{N1} and F_{N2}, and the magnitude of friction is $F_{Ni} = U_N F_{Ni}$, where the friction coefficient U_N is the normal friction coefficient.

Table 1
Three environmental scenarios.

Material	Friction coefficients (with robot)	Size (m)	Depth (mm)
Grass	$U_t = 0.79, U_n = 1.18$	$4 * 3$	8
Sand	$U_t = 0.47, U_n = 0.71$	$3 * 2$	5
Sponge	$U_t = 0.91, U_n = 1.03$	$3 * 3$	10

4. Experiments

Three environmental scenarios were designed to assess the movement of the snake robot. These scenarios represent two outdoor environments, namely grassy terrain and sandy terrain, and one indoor environment characterized by a foam mat surface. During the experiments, markers were affixed to the snake robot, and an external vision system, NOKOV, was employed to record the data of motion. The results indicated that although there were some variations in the performance of these five gaits across different environmental scenarios, the snake robot's motion was not significantly impacted by changes in the environment. The robot demonstrated effective adaptability to each scenario as shown in Table 1.

4.1. Sidewinding experiment

The experimental results of the sidewinding state are presented in Fig. 10, depicting the snake robot's motion across three environmental scenes: grass, sand, and a sponge mat (shown in Fig. 9). In this gait, the primary movement of the snake robot involves lateral displacement along the X-axis, with a minor displacement in the Y-axis direction. However, the fluctuation amplitude of Vx significantly surpasses that of Vy. The snake robots exhibit swift sideways motion, characterized by high-speed gaits. The average speeds of Vx on grassland, sand, and the sponge mat were 20.15 mm/s, 14.35 mm/s, and 18.74 mm/s, respectively, while the average speeds of Vy on grassland, sand, and the sponge mat were 2.48 mm/s, −1.55 mm/s, and −0.53 mm/s, respectively. The acceleration of the snake robot in the Y direction was utilized to calculate the frictional force acting on the snake robot.

4.2. Rolling experiment

In the rolling experiments depicted in Fig. 11, the snake robot's motion state is recorded in Fig. 12 across three distinct environmental surfaces: grass, sand, and foam mats. During the rolling gait, one side of the snake robot contacted with the ground and acted as the pivot axis for the rolling motion. The characteristics of rolling motion on grass, sand, and foam mats exhibit similarities, with the lateral rolling gait primarily relying on ground reaction forces and experiencing minimal influence from

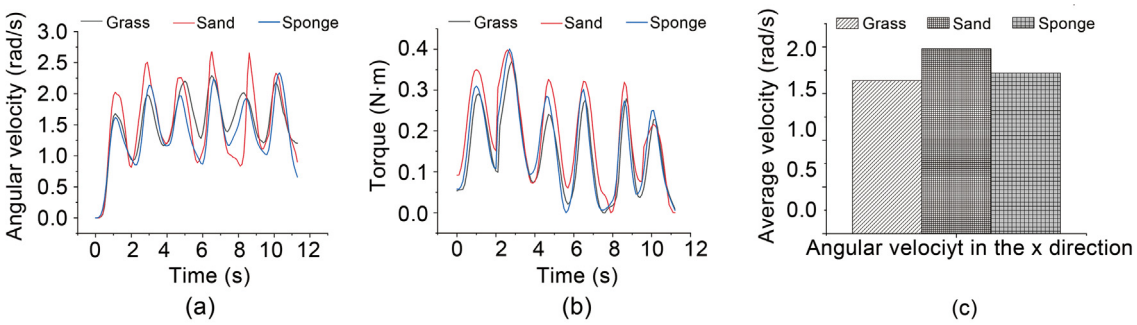


Fig. 12. Motion state of rolling.

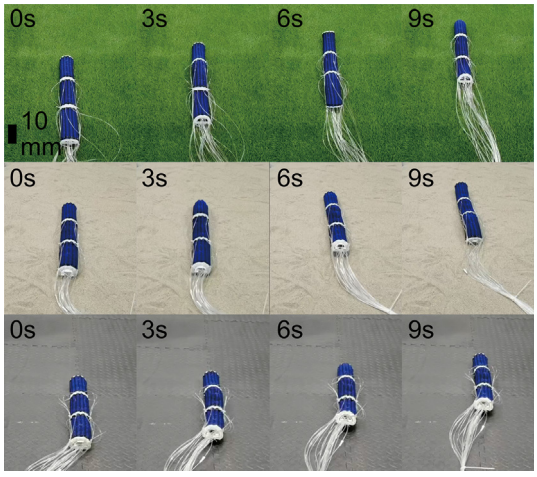


Fig. 13. Longitudinal traveling wave gait in three scenarios.

friction. The average rolling velocities of the snake robot on grass, sand, and foam mats are 1.64 rad/s, 1.98 rad/s, and 1.72 rad/s, respectively. Additionally, we computed the torque generated by the snake robot during lateral rolling gait based on its angular acceleration.

4.3. Longitudinal traveling wave experiment

In the longitudinal traveling wave gait experiment shown in Fig. 13, the motion state of the snake robot is recorded in Fig. 14 across three environmental scenarios: grass, sand, and sponge mat. In the longitudinal traveling wave gait, the snake robot exhibits inchworm-like motion along the *Y*-axis, with its motion speed fluctuating in tandem with the stretching frequency of the inchworm motion. Notably, on sand, the snake robot's speed in the longitudinal traveling wave decreases progressively due to the reduced friction coefficient as it moves and displaces sand. The average moving speed of the snake robot on grass, sand, and foam mats was 7.74 mm/s, 2.7 mm/s, and 5.14 mm/s, respectively. Additionally, we computed the force of friction acting on the snake robot in the *Y* direction based on the robot's acceleration.

4.4. Gait movement characteristics

The distinctive characteristics of these five gaits are summarized as follows:

Sidewinding Gait: This gait allows lateral movement while introducing some forward and backward displacement. It is characterized by rapid lateral motion, and when faced with obstacles, its ability to navigate around them passively is facilitated by the combined lateral and longitudinal movements.

Rolling Gait: The lateral rolling gait involves lateral axial rolling and requires a relatively large motion space. When confronted with obstacles, it can actively overcome them by rolling over, and it relies on ground reaction forces, resulting in minimal impact from environmental changes.

Longitudinal traveling: This gait imitates inchworm-like movement, enabling slow but efficient forward and backward motion. It requires less spatial clearance and can navigate through narrow environments.

These characteristics provide a comprehensive understanding of the adaptability and utility of each gait in different scenarios and environments.

4.5. Complex environments

Finally, we conducted experiments with the snake robot in a complex environment, as illustrated in Fig. 15. This intricate setting encompassed grassy terrain, wooden stick obstacles (0.5 m in length and 12.5 mm in diameter), narrow pipe passages (inner diameter of 100 mm), sandy terrain, and a straight track scene. This experimental arrangement emulates the challenging

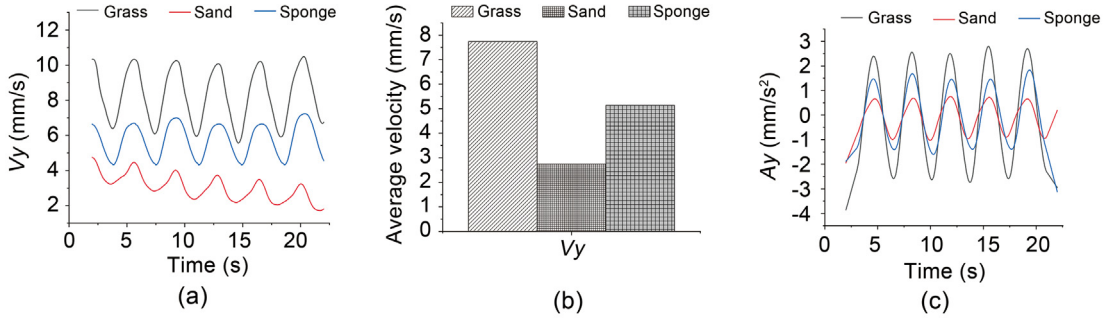


Fig. 14. Motion state of longitudinal traveling wave.



Fig. 15. Complex environments of the snake robot.

outdoor environments typical of real snakes. Moreover, the scenarios underwent frequent changes, requiring the snake robot to demonstrate a diverse range of locomotion capabilities to navigate environmental shifts.

In the initial 30 s, the snake robot employs the lateral wave gait as it navigates through the grass scene, eventually reaching the stick obstacle scene. Encountering an obstruction at the front, the snake robot, between 40 and 70 s, transitions to lateral movement gait, repositioning itself to the lower-left side of the stick obstacle scene. Subsequently, the snake robot adjusts its posture and executes the rolling gait to surmount these obstacles. After 15 s, the snake robot successfully conquers the stick obstacles, progressing to the narrow pipeline scene in 1 min and 30 s. In the narrow pipe scene, characterized by limited entrance space, the snake robot, from 1 min and 55 s to 2 min and 55 s, approaches and enters the narrow pipe using straight-line motion with a longitudinal traveling wave gait. Although confined within the narrow pipe, the longitudinal traveling wave gait allows the snake robot to generate propulsive force by rubbing against the inner wall. After sustaining this motion for a period, the snake robot successfully traverses the narrow pipe, arriving at the sandy scene by 8 min and 50 s. Between 9 min and 30 s and 10 min, the snake robot approaches the linear orbit scene using lateral moving gait. Due to most of the snake robot's body remaining in the narrow pipe, resulting in minimal tail displacement during lateral movement, the snake robot extends its tail through the narrow pipe in the sand scene between 10 and 10 min and 30 s. From 10 min and 30 s to 10 min and 50 s, the snake robot executes lateral tumbling gait as it approaches the linear track scene. Between 10 min and 50 s and 12 min and 30 s, the snake robot adjusts its posture using both the longitudinal traveling wave gait and lateral moving gait until it enters the straight track. From 13 min and 10 s to 17 min and 40 s, the snake robot first enters the linear orbit using longitudinal traveling wave gait and then departs from

the linear orbit through linear stretching gait. By 20 min and 40 s, the snake robot is on the verge of completely leaving the straight track, providing ample space for other gaits. At this point, around 21 min, the snake robot executes a lateral rolling gait and flips over the straight track. Ultimately, after 21 min, the snake robot successfully completes the traversal of the entire complex environment.

Throughout the experiment, the snake robot adeptly adjusts its motion gait in response to diverse environmental scenes, showcasing its capability to modulate spatial positioning through a repertoire of moving gaits. This underscores the advantages of a snake robot equipped with multiple locomotion gaits, emphasizing its adaptability and agility in navigating complex environments.

5. Conclusion

To enhance the locomotion and environmental adaptability of the snake-like robot, we introduced a soft-bodied snake robot featuring a diverse range of locomotion gaits. Comprising three interconnected soft actuators representing the head, body, and tail segments, this robot incorporates radially arranged pneumatic artificial muscles at the bottom of each soft actuator. This design introduces anisotropic friction during motion, and enabling versatile movement. The soft actuators exhibit flexibility in both horizontal and vertical directions, along with radial extension, forming the basis for three-dimensional gaits in the snake robot.

The developed robot showcases an extended range of locomotion patterns, surpassing the capabilities of previous single-mode robots [37,38]. It seamlessly executes sidewinding, rolling, and longitudinal traveling wave motions, significantly enhancing its adaptability to diverse environments. Whether maneuvering through regular grassy terrain, traversing loose sandy ground, or gliding across soft sponge padding, the robot excels with these three locomotion modes. Noteworthy is the robot's exceptional performance in complex environments, where it adeptly overcomes obstacles, navigates narrow passages, and operates efficiently in confined spaces. The developed soft-bodied snake robot demonstrated its motion capability and adaptability across a wide spectrum of environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.birob.2024.100148.

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TECHNICAL REVIEW · SOFT ROBOTICS

Soft Robotic Snakes

Architectures, actuation, modelling, control and open problems in continuum serpentine locomotion

A survey of the soft snake robotics field

Compiled August 2026

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1 Introduction and Scope

ABSTRACT

Snake robots occupy a distinctive niche in robotics: they are hyper-redundant, they can enter spaces no wheeled or legged machine can reach, and they generate propulsion without any dedicated propulsive appendage. The field divides into two lineages that have developed largely in parallel. The *rigid lineage*, beginning with Hirose's Active Cord Mechanism in 1972, comprises multi-link chains of motorised joints, and is by far the more mature: it has produced field-deployed systems for subsea inspection, disaster response and planetary exploration. The *soft lineage*, roughly two decades old, replaces discrete joints with continuously deformable elastomer bodies actuated by pneumatic chambers, artificial muscles or electroactive polymers. Soft snakes are slower, weaker and harder to model, but they are inherently safe, mechanically robust to impact and constriction, and capable of conforming to unstructured terrain in ways rigid chains cannot. This review surveys the soft lineage: what has been built, how it is actuated, how it is modelled and controlled, what it can currently achieve, and where the field's binding constraints lie.

1.1 Why softness

The case for soft snake robots rests on four properties that follow from continuous compliance rather than being engineered in.

Conformability. A continuum body distributes contact along its entire length rather than at discrete joints. In cluttered environments this converts obstacles from obstructions into anchors — the phenomenon of *obstacle-aided locomotion*, where a snake pushes off irregularities to generate thrust that would be unavailable on smooth ground.

Intrinsic safety. Elastomer bodies with moduli in the 10 kPa to 1 MPa range cannot generate the contact forces that make rigid manipulators dangerous. This matters for medical applications and for search-and-rescue in collapsed structures where a rigid machine could destabilise debris.

Impact and constriction tolerance. A cast silicone body has no bearings to jam, no gearboxes to strip, and can be crushed and recover. Rigid snake robots fail at joints; soft snakes largely fail at seams and electrical interfaces.

Simplicity of construction. A soft snake segment can be a single moulded part. The mechanical parts count of a soft snake is typically an order of magnitude below its rigid equivalent.

1.2 The countervailing costs

These advantages are purchased at real cost, and the field's literature is candid about it.

- **Infinite-dimensional configuration space.** A continuum body has no finite set of joint angles. Every model is a truncation, and the truncation error is a first-class engineering concern rather than a rounding detail.
- **Underactuation.** Soft snakes typically have far fewer independent actuators than deformation modes, and the body pose is never directly actuated at all.
- **Hysteresis and rate dependence.** Elastomers are viscoelastic. Actuator response depends on history, not just on the current command.
- **Power.** Pneumatic snakes need compressors; electroactive snakes need kilovolts. Neither miniaturises easily, and untethered operation remains the field's most persistent unsolved problem.
- **Speed.** The fastest controllable soft snakes reach roughly one body length per second. Rigid snakes and most legged robots comfortably exceed this.

2 Biological Foundations

Snake robotics is unusual among bioinspired fields in that the biology is not merely motivational — the quantitative biological results are directly load-bearing in robot design. Three findings in particular determine whether a snake robot works at all.

2.1 Frictional anisotropy is the propulsion mechanism

A snake has no limbs, no wheels and no propeller. Forward motion comes entirely from the difference between how easily the body slides along its own axis versus across it. Hu, Nirody, Scott and Shelley established the governing measurements in 2009: for live snake skin on a smooth surface, friction is lowest in the forward direction, roughly twice as high laterally, and roughly four times as high backwards.

```
Anisotropy ratios (biological baseline)

    forward  :  lateral  :  backward
       1     :     2     :      4

    η ≡ μ_lateral / μ_forward  ≈  2
```

The consequence for robotics is severe and frequently underappreciated. It can be shown that with *isotropic* friction, a free-floating snake executing any periodic shape change produces zero net displacement over a cycle — the body wiggles and the centre of mass stays put. Anisotropy is not an enhancement to locomotion; it is the entire mechanism.

> **DESIGN CONSEQUENCE**
>
> Skin design deserves as much engineering effort as actuator design. A soft snake whose belly is bare cast silicone will typically show `η ≈ 1.1–1.3` and will barely translate regardless of control sophistication. Measure the anisotropy ratio on a tribometer before committing to a full build; target `η ≥ 2`, ideally 3–5, via moulded directional scales, angled bristles or a printed anisotropic texture.

2.2 The gait repertoire

Herpetology recognises four locomotor primitives, and essentially all robot gaits derive from them or from combinations.

Gait	Mechanism	Robot relevance
Lateral undulation	Travelling lateral wave; whole body in continuous ground contact; thrust from lateral friction reaction	The default soft-snake gait. Requires anisotropic friction and open ground
Sidewinding	Superposed horizontal and vertical waves with a phase offset; body rolls like an elliptical tread; static contact patches	Excellent on loose granular slopes. Requires out-of-plane actuation — rare on soft snakes
Concertina	Alternating anchoring and extension of body sections	Confined channels and tunnels. Energetically very costly
Rectilinear	Ventral scale waves; body remains straight	Slow, discreet, works in narrow tubes. Suits peristaltic soft designs

To these robotics adds helical rolling for pole and pipe climbing, obstacle-aided locomotion in cluttered terrain, anguilliform and carangiform swimming, sinus-lifting, and burrowing in granular media.

2.3 Scale control and dynamic weight redistribution

Real snakes actively modulate their belly scales and, more significantly, lift regions of high body curvature off the ground during undulation. This redistributes the normal load toward body segments that are generating useful thrust and away from those that are merely being dragged sideways. The measured speed improvement is substantial.

Almost no soft snake robot implements this. It requires either out-of-plane actuation or a variable-stiffness mechanism, and it represents a well-identified, quantitatively motivated, and largely unexploited performance opportunity.

3 Taxonomy and Landmark Platforms

3.1 The rigid lineage: context for the soft field

Shigeo Hirose built the first serpentine robot, the Active Cord Mechanism ACM-III, in 1972: roughly two metres long, twenty joints, passive caster wheels providing the frictional anisotropy that biological scales provide in snakes. Hirose also formalised the *serpenoid curve*, the shape in which curvature varies sinusoidally along the body, which remains the parametric basis of nearly every lateral-undulation gait implemented since. His 1993 monograph *Biologically Inspired Robots* is still the field's foundational text.

The lineage continued through ACM-R3 (three-dimensional mobility via alternating pitch and yaw joints), ACM-R5 (amphibious, sealed), and the Souryu rubble-search crawlers. In parallel, Carnegie Mellon's Biorobotics Lab produced the modular snake robot family — sixteen single-degree-of-freedom modules, roughly 94 cm long and 5 cm in diameter — demonstrating pole climbing by helical rolling, sidewinding on granular slopes, swimming, and archaeological survey. NTNU's group produced the most complete body of modelling and control theory in the field, along with the Kulko, Wheeko and Mamba platforms and the underwater work that became Eelume.

3.2 Soft and continuum platforms

The reference soft snake platform is the series developed at Worcester Polytechnic Institute by Cagdas Onal's group. Each module comprises two silicone bending chambers arranged antagonistically about a central inextensible constraint layer, with fibre reinforcement limiting radial ballooning. Pressurising one chamber bends the module; alternating pressurisation along a chain of modules produces a travelling wave. The second-generation platform reached approximately 220 mm/s, about one body length per second and roughly ten times the first generation, making it one of the fastest controllable soft mobile robots reported. Later work added embedded Hall-effect curvature sensors, sliding-mode valve control, and reinforcement-learning gait regulation.

Other significant threads include electroactive soft swimmers using antagonistic dielectric elastomer actuators in laminated silicone; magnetically steered continuum devices for endoluminal medicine; growing or eversion "vine" robots that extend from the tip by pressure-driven inversion; and origami-based serpentine designs that trade some compliance for predictable folding kinematics.

3.3 Hybrid and origami architectures

A pragmatic middle ground places rigid electronics, valves and batteries in discrete pods connected by soft actuated segments. This resolves the awkward fact that soft bodies are poor hosts for the stiff components a robot needs. Origami serpentine designs use folded patterns to obtain large, repeatable deformation from modest actuation, at the cost of anisotropic and somewhat brittle mechanics.

3.4 Commercial and field-deployed systems

System	Origin	Domain
Eelume	NTNU spinout	Resident subsea inspection, maintenance and repair
Sarcos Guardian S	Sarcos Robotics	Magnetic-tread inspection of pipes and confined spaces
OC Robotics snake-arm	UK, later GE	Confined-space aerospace manufacturing, nuclear inspection
Medrobotics Flex	CMU spinout	Transoral and transanal surgery; FDA-cleared 2015
NASA JPL EELS	JPL with CMU, ASU, UCSD	Icy-moon and glacier exploration; ~100 kg, ~4 m, screw-propelled

Note that none of these are soft robots. The soft lineage has not yet produced a fielded system, which is itself an informative statement about the field's maturity.

4 Locomotion Gaits and Their Mathematics

4.1 The serpenoid curve

Expressed as a travelling curvature wave along the arc length s:

$$\kappa(s,t) = \alpha \cdot \sin(2\pi\, s / \lambda - \omega\, t) + \kappa_0$$

α	curvature amplitude	[1/m]
λ	spatial wavelength	[m]
ω	temporal frequency	[rad/s]
κ_0	curvature bias → steering	[1/m]

Discretising onto N modules of length $\ell = L/N$ requires averaging the continuous curvature over each module, which introduces a factor frequently omitted:

$$\kappa_i(t) = \alpha \cdot \mathrm{sinc}(\pi\, \ell / \lambda) \cdot \sin(\omega\, t - 2\pi\, (i - \tfrac{1}{2})\, \ell / \lambda) + \kappa_0$$

where $\mathrm{sinc}(x) = \sin(x)/x$

The naive substitution without the sinc factor over-drives the modules. The error exceeds 10% at four modules per wavelength. In practice at least six to eight modules per wavelength are needed for the discretised wave to resemble the continuous serpenoid, which sets a hard lower bound on module count for any robot intended to execute a full body wave.

4.2 Gait parameters and their effects

Parameter	Physical effect	Typical range	Limited by
α	Lateral excursion and hence thrust per cycle	5–30 m^{-1}	Actuator stroke
λ	Number of body waves; affects stability and turning	0.5L – 1.5L	Module count
ω	Speed, roughly linearly until saturation	0.5–5 Hz	Actuator bandwidth
κ_0	Turning; radius approximately $1/\kappa_0$	0–5 m^{-1}	Curvature limit

A caveat on the classical model. The conventional serpenoid assumes fixed parameters. Changing them mid-motion — to optimise speed or avoid an obstacle — introduces discontinuities in the commanded shape that cause segment slip and loss of thrust. Modified serpenoid formulations with smoothly varying parameters address this and should be preferred for any robot doing real-time gait adaptation.

4.3 Gaits available and unavailable to soft bodies

Soft snakes execute lateral undulation naturally, rectilinear and peristaltic gaits well, and concertina passably. They struggle with sidewinding and helical rolling, both of which require coordinated out-of-plane bending and, more importantly, the ability to hold shape against gravity while parts of the body are unsupported. A soft body with low bending stiffness sags. This is the clearest current capability gap between soft and rigid snakes, and it is a stiffness problem rather than an actuation problem — which is why variable-stiffness mechanisms are of disproportionate interest.

5 Actuation Technologies

Actuator choice determines nearly everything downstream: bandwidth, force, power architecture, modelling approach and whether untethered operation is feasible. It is the first decision and the hardest to reverse.

5.1 Pneumatic actuation

Fibre-reinforced elastomer actuators and PneuNets dominate the soft snake literature. A chamber cast into an elastomer inflates; an inextensible strain-limiting layer on one side converts expansion into bending; helical fibre wrapping prevents radial ballooning and directs deformation axially. Typical operating pressures for Ecoflex-class bending actuators are 10–60 kPa, with large bending angles achievable at the lower end of that range.

Strengths: high force, large stroke, well-understood fabrication, inherently compliant, tolerant of overload. **Weakness:** the supply. No compressor exists that is simultaneously small enough, quiet enough and powerful enough for an untethered soft snake of practical size. Nearly all pneumatic soft snakes are tethered, and this single constraint has arguably shaped the field's trajectory more than any other.

5.2 Dielectric elastomer actuators

A DEA is a compliant capacitor: an elastomer film between stretchable electrodes. Applied voltage generates Maxwell stress that compresses the film through its thickness and expands it in area.

```
p = ε₀ · ε_r · (V / h)²

    ε₀  = 8.854e-12 F/m
    ε_r ≈ 2.8–3.5 for silicone, ≈ 4.7 for acrylic
    h   = film thickness
```

Three consequences follow directly from this expression and govern all DEA design. First, force scales with the *square* of voltage, so blocking force curves are parabolic. Second, force scales with the *inverse square* of thickness, so halving film thickness quarters the voltage needed for equal stress — this is why multilayer thin-film construction is the central voltage-reduction strategy. Third, the stress is independent of voltage sign: a DEA can push but never pull, which is precisely why antagonistic pairs are required for bidirectional bending.

Materials. Acrylics (3M VHB) achieve extraordinary strains — linear above 380%, areal above 2000% under high pre-strain — but are strongly viscoelastic and therefore slow. Silicones (Sylgard, Ecoflex, Elastosil) are faster with far less hysteresis but deliver lower strain.

Pre-stretch raises breakdown strength and suppresses electromechanical pull-in instability, but conventionally requires a rigid frame — which is fundamentally at odds with a monolithic soft body. Resolving this tension, via embedded strain-limiting spines or no-pre-stretch multilayer chemistries, is an active research question.

Voltage. Historically kilovolts. Conventional silicone DEAs at 50 µm thickness require 4–5 kV for modest force. Recent work has cut this dramatically: fluorinated dielectric formulations achieving large strain at low field, and multilayer designs operating untethered at a few hundred volts, have appeared in the last two years. This is the most rapidly moving area in the field.

5.3 HASEL and electrohydraulic actuators

Hydraulically amplified self-healing electrostatic actuators use electrostatic zipping of electrode pairs to displace a liquid dielectric, producing muscle-like linear contraction. The liquid dielectric self-heals after breakdown, removing DEA's catastrophic failure mode. Peano-HASEL variants contract roughly 15% as standard and up to about 24% in high-strain designs. Classical HASELs need in excess of 6 kV; low-voltage variants using high-permittivity shells and multi-chamber geometries have reduced this toward the kilovolt range.

5.4 Shape memory alloys

Nickel-titanium wire or coil contracts on Joule heating. Antagonistic pairs give bidirectional motion. SMAs offer very high force density and trivially simple construction. The binding constraint is thermal: heating is fast but cooling is slow and passive, capping practical bandwidth near 0.5 Hz. Efficiency is below 5%. Embedding SMA in silicone compounds the problem, since the elastomer insulates. Reactivating before full cooling risks permanent alteration of the alloy's crystal structure. Control is normally closed-loop on wire electrical resistance, since direct temperature measurement on sub-millimetre wire embedded in elastomer is impractical.

5.5 Tendon drive

Cables routed along a compliant backbone, actuated by motors at the base. Common in continuum manipulators and surgical devices. Advantages: mature, powerful, all electronics remote. Disadvantages: friction and backlash in the routing accumulate along the body, and the actuation is not distributed — a long tendon-driven snake becomes progressively harder to control toward the tip.

5.6 Magnetic actuation

Magnetic particles embedded in the elastomer respond to external field gradients. Dominant in medical continuum robotics because it removes all onboard actuation and power. Limited to environments where an external field can be imposed, so irrelevant to field robotics but the leading approach for endoluminal devices.

5.7 Comparison

Technology	Drive	Strain	Force	Bandwidth	Untethered?	Principal limitation
Pneumatic FEA	10–60 kPa	Large bending	High	1–5 Hz	Very difficult	No adequate onboard compressor
DEA	0.2–5 kV	10–100%+	Low–mid	10–100+ Hz	Emerging	High voltage; pre-stretch needs frame
HASEL	1–8 kV	15–25%	Mid–high	10–50 Hz	Emerging	High voltage; liquid containment
SMA	4–6 V, ~1 A	4% wire, 200% coil	Very high	0.2–0.5 Hz	Yes	Cooling-limited; <5% efficient
Tendon	Motor	Geometry-set	High	5–20 Hz	Yes	Friction accumulates along body
Magnetic	External field	Large bending	Low	1–10 Hz	N/A	Requires external field source

6 Materials and Fabrication

6.1 Elastomer selection

Material	Shore	Elongation at break	Typical role
Ecoflex 00-10 / 00-30	00-10 / 00-30	~800–1000%	Highly compliant bending chambers; low actuation pressure
Ecoflex 00-50	00-50	~980%	Slightly stiffer chambers; better shape retention
Dragon Skin 10 / 20 / 30	10A–30A	~1000%	Structural bodies, skins, higher-pressure chambers
Sylgard 184	~43A	~120%	DEA dielectric films; stiff, optically clear, low loss
Elastosil (film grades)	Varies	Varies	Precision thin films for DEA multilayers

Softer elastomers require less actuation effort but retain shape poorly and sag under gravity. Stiffer elastomers hold gait shape but demand more pressure or voltage. Most designs use a graded approach: soft chambers, stiffer structural spine, and a tough skin.

6.2 Constitutive models

Linear elasticity is invalid above roughly 20% strain, which soft snakes routinely exceed. Hyperelastic strain-energy formulations are required.

Model	Character	When to use
Neo-Hookean	One parameter	Only for small strain; poorest fit at large strain
Mooney-Rivlin	Two parameters	Moderate strain, historical convention
Yeoh (3rd order)	Three parameters, I_1 only	Best choice when only uniaxial test data is available
Ogden (3rd order)	Six parameters, principal stretches	Most accurate overall; needs multi-mode test data
Gent	Two parameters, limiting stretch	Captures strain stiffening; essential for DEA pull-in analysis

The comparative study by Marechal and colleagues, covering seventeen elastomers under ASTM D412, is the standard reference and provides fitted coefficients for common Ecoflex and Dragon Skin grades directly.

6.3 Fabrication routes

- **Mould casting** — dominant for monolithic bodies. Multi-part moulds with soluble or lost-wax cores produce internal chambers. Cheap, reliable, low resolution.
- **Direct ink writing and embedded 3D printing** — deposits uncured elastomer into a support matrix. Enables complex internal channel geometries impossible to mould.
- **Spin coating** — for DEA thin films. Achieves tens of microns before pre-stretch, single microns after.
- **Fibre reinforcement** — helical winding of inextensible thread constrains radial expansion and directs deformation.
- **Strain-limiting layers** — a stiff inextensible sheet bonded off-axis converts symmetric expansion into bending. The single most important fabrication trick in soft robotics.

6.4 Anisotropic skins

Given the argument of Section 2.1, the skin is a primary functional component rather than a covering. Approaches include moulded directional scales that catch on backward sliding, angled bristle arrays, printed or laser-textured surfaces, and passive wheels for robots willing to sacrifice full softness. Report the measured ratio, not the intended one — moulded scales frequently underperform their design values because the elastomer's own compliance lets them fold flat under load.

7 Modelling

Soft snake modelling operates on a spectrum from fast and approximate to faithful and slow. Mature practice uses two models simultaneously: a reduced model for control synthesis, and a continuum model for physical fidelity, with the former calibrated against the latter and the latter validated against hardware.

7.1 The kinematic layer

All models factor through three spaces: actuator space (pressures or voltages), configuration space (the curvature field), and task space (backbone pose). The map from configuration to task space is purely geometric and well-conditioned. The map from actuator to configuration space contains all the electromechanics, hyperelasticity and hysteresis, and is where nearly all modelling error originates.

Piecewise constant curvature (PCC) assumes each module bends as a circular arc. The transform across a module of length ℓ and curvature κ:

```
       [ cos(κℓ)   −sin(κℓ)    sin(κℓ)/κ    ]
  T  =   [ sin(κℓ)    cos(κℓ)   (1−cos(κℓ))/κ ]
       [    0          0            1       ]
```

> **IMPLEMENTATION WARNING**
>
> Both `sin(κℓ)/κ` and `(1−cos(κℓ))/κ` are indeterminate at zero curvature, and straight configurations occur twice per module per gait cycle. Guard with series expansions below `|κℓ| < 1e-4`. This single detail accounts for more debugging time in continuum robotics than any other.

PCC is exact for a uniformly actuated segment under no external load, and progressively wrong as loading becomes non-uniform — which, during ground contact, it always is.

7.2 Lagrangian dynamics of a discretised body

Taking generalised coordinates as the base pose plus module curvatures, the Euler-Lagrange equations produce the familiar manipulator form:

```
M(q) q̈  +  C(q,q̇) q̇  +  K(q − q*)  +  D q̇  =  Jᵀ F_friction  +  B τ_actuation
```

Three structural properties carry over from rigid manipulator theory and are worth exploiting: the mass matrix is symmetric positive definite; $\dot{M} - 2C$ is skew-symmetric, which supplies a free Lyapunov function for stability proofs; and the dynamics are linear in the inertial parameters, which enables both adaptive control and least-squares system identification.

The input matrix has zero rows corresponding to the base pose. The robot is underactuated by exactly three degrees of freedom in the plane, and every metre of forward progress is produced by the coupling term $J^{\mathsf{T}} F_{\text{friction}}$. Set friction to zero and the centre of mass provably cannot move.

7.3 Cosserat rod theory

The geometrically exact continuum model. The rod is a curve of material frames; the state comprises position, orientation, internal force and internal moment as functions of arc length and time.

```
Kinematics
    ∂p/∂s = R v          ∂R/∂s = R û
    ∂p/∂t = R q          ∂R/∂t = R ŵ

Compatibility
    ∂v/∂t = ∂q/∂s + u × q − w × v
    ∂u/∂t = ∂w/∂s + u × w

Balance
    ∂n/∂s + f = ρA ∂²p/∂t²
    ∂m/∂s + (∂p/∂s) × n + l = R( ρJ ∂w/∂t + w × ρJ w )

Constitutive
    n = R K_se (v − v*)      K_se = diag(GA, GA, EA)
    m = R K_bt (u − u*)      K_bt = diag(EI, EI, GJ)
```

Actuation enters most cleanly through the reference strain u^* — the *spontaneous curvature*. Rather than applying point moments, an actuator changes the rod's stress-free shape. This formulation underpins the theoretical result that controlling spontaneous curvature under anisotropic frictional constraint generates locomotion, and it is the natural home for any distributed actuator.

A free-crawling snake has free-free boundary conditions, which makes the spatial boundary-value problem singular up to rigid-body motion. The standard remedy is to fix a reference frame at one material point, integrate in that frame, and recover global motion from momentum balance.

For planar robots the system reduces to a scalar partial differential equation in the tangent angle — the *active elastica* — which captures everything a planar snake does at a small fraction of the computational cost of the full three-dimensional rod.

7.4 Discrete Cosserat and piecewise constant strain

Renda and colleagues discretise the continuous Cosserat model by assuming piecewise constant *strain* rather than curvature on each section. This permits analytic spatial integration and extends PCC to include shear and torsion at essentially no additional cost, while yielding a formulation with

direct structural parallels to rigid manipulator theory — the same mass, Coriolis and stiffness matrix structure, with a rigorous continuum underpinning.

> For a *crawling* robot this is more than an elegance argument. Shear and torsional deformation are essential to represent out-of-plane external loads, and ground contact loads are precisely out-of-plane. Piecewise constant strain is therefore the recommended default for soft snake locomotion, over piecewise constant curvature.

7.5 Friction modelling

Decompose each contact element's sliding velocity into body-frame tangential and normal components and apply direction-dependent coefficients. Coulomb friction with a signum function makes the system discontinuous, which breaks most integrators and all gradient-based optimisers. Standard practice regularises:

```
sgn(v)  ≈  tanh( v / v_ε )
```

The choice of `v_ε` is a modelling decision, not a numerical detail. Too large and viscous damping is silently introduced and thrust under-predicted; too small and the system becomes numerically stiff. A defensible practice is to set it near 1% of the characteristic body-point speed, sweep it over a decade either side, and report the resulting sensitivity of predicted locomotion speed.

Where stick-slip transitions matter, LuGre models with an internal bristle state, or nonlinear complementarity formulations smoothed to equalities, provide higher fidelity at greater computational cost.

7.6 Geometric mechanics

The most powerful framework for gait *design*, as opposed to simulation. Separating the configuration into shape variables and position variables, drag-dominated systems obey a linear relation between shape velocity and body velocity via the *local connection*. Net displacement over a closed gait loop then follows, by Stokes' theorem, as the area integral of a scalar *height function* over the region the gait encloses in shape space.

This converts gait optimisation from black-box search over amplitude, wavelength and frequency into a two-dimensional area-maximisation problem with visual interpretation. It also explains why gait ordering matters: the Lie bracket term in the height function encodes non-commutativity, so sequences that appear equivalent are not. A practical subtlety is that the approximation error depends on the choice of body frame, and minimum-perturbation coordinates should be used for large-amplitude gaits.

7.7 Data-driven and reduced-order models

Proper orthogonal decomposition on simulated or measured curvature fields typically shows that two modes capture over 95% of the energy for lateral undulation — the sine and cosine pair that constitute a travelling wave. This provides quantitative justification for the two-dimensional shape-space reductions used in geometric mechanics.

Koopman operator methods lift the state into an observable space where dynamics are approximately linear, enabling linear MPC on a nonlinear soft robot. The approach is data-hungry, extrapolates poorly outside its training distribution, and handles frictional non-smoothness badly. It is best regarded as an accelerator for real-time control rather than a model of record.

8 Sensing and State Estimation

A rigid robot knows its shape from joint encoders. A soft robot has no joints, and shape sensing is correspondingly harder — arguably the field's second unsolved problem after power.

Method	Principle	Assessment
Magnetic (Hall-effect)	Magnet and sensor on a flexible substrate in the neutral axis; field varies with bending	Proven on the WPI platform; robust, low cost; cross-sensitivity to external fields
Resistive / liquid metal	Conductive channel resistance changes with strain	Simple, fully soft; drift and hysteresis are significant
Capacitive	Parallel-plate capacitance changes with deformation	Low hysteresis; sensitive to environmental capacitance
Fibre Bragg grating	Wavelength shift under strain, distributed along a fibre	Excellent accuracy and distributed measurement; interrogator is expensive and bulky
IMU chains	Orientation at discrete points, integrated to shape	Mature; drifts without correction; adds rigid mass
DEA self-sensing	Actuator capacitance is a function of stretch	Zero added mass; only available with electroactive actuation

Self-sensing deserves emphasis. Since a dielectric elastomer actuator's capacitance varies with its stretch, superimposing a small high-frequency probe signal on the drive voltage recovers local strain from the actuator itself. For an antagonistic pair, the difference in measured stretch gives local curvature directly. Every actuator becomes its own sensor at no mass penalty — a substantial practical advantage in a domain where embedding sensors is genuinely difficult.

9 Control

9.1 Central pattern generators

Coupled nonlinear oscillators generate the rhythmic activation underlying gait, mirroring the neural architecture of biological locomotion. Matsuoka oscillators, built from mutually inhibiting neuron pairs with adaptation, map naturally onto antagonistic actuator pairs — the flexor and extensor outputs drive the two sides directly. Hopf oscillators and Kuramoto phase coupling are common alternatives. Adjacent modules are coupled with a phase lag, producing the travelling wave.

CPGs offer smooth trajectories by construction, graceful gait transitions, independent modulation of amplitude, frequency and phase, and a small parameter set amenable to learning.

9.2 Why open-loop CPG plus PID disappoints

CPG-generated trajectories presuppose competent low-level tracking. Soft actuators do not track well: their response is nonlinear, delayed, hysteretic and somewhat stochastic. The result is that a controller which behaves beautifully in simulation degrades badly on hardware. Documented remedies include model reference adaptive control, which forces heterogeneous soft actuators to behave like a common reference model; feedback linearisation with unscented Kalman filtering for state estimation; sliding-mode control driving binary valve states directly rather than through pulse-width modulation, which allows position holding without continuous pressurisation; and robust control with explicit anti-windup, which matters because actuator inputs saturate hard at breakdown voltage or maximum pressure.

9.3 Reinforcement learning

The most productive recent direction fuses learning with CPGs rather than replacing them. Rather than learning joint trajectories from scratch, a learned policy regulates the tonic inputs of a CPG, which retains rhythmicity and smoothness by construction while the policy handles goal-directed adaptation. This has been demonstrated end-to-end on soft snake hardware with steering and velocity control, including sim-to-real transfer. The architecture is deliberately biomimetic: a learned "brain" modulating a fixed "spinal cord."

Sim-to-real transfer remains difficult. Domain randomisation over friction coefficients, actuator time constants and material parameters is standard practice; the friction parameters are usually the ones that matter most.

9.4 Path following and guidance

For directed locomotion rather than pure gait generation, the mature approach is line-of-sight guidance with integral action, layered above a sinusoidal gait generator and a heading controller. This body of work — developed largely for rigid and underwater snakes — transfers to soft platforms wherever heading can be estimated reliably.

10 Power and Electronics

Power is the field's binding constraint. Every soft snake capability discussion eventually reduces to it.

10.1 Pneumatic supply

Tethered operation with an external compressor is standard, and the tether imposes drag, snagging and range limits that materially affect experimental results. Onboard alternatives — miniature diaphragm pumps, compressed gas cartridges, chemical gas generation, phase-change fluids — each fail on some combination of flow rate, duration, mass or controllability. No general solution exists.

10.2 High-voltage drive for electroactive actuators

Electroactive snakes trade the compressor problem for the kilovolt problem, which is more tractable because it is an electronics problem rather than a thermodynamics one. Progress has been rapid. Miniature boost and flyback topologies now deliver hundreds of volts to several kilovolts from single-cell lithium inputs at masses measured in hundreds of milligrams to a few grams. Ultra-high-gain converter families reaching conversion ratios near a thousand at efficiencies approaching 80% have been demonstrated specifically for soft robotic use. Switching is handled by high-voltage optocouplers, which are fast and isolated, or by series MOSFET stacks, which are lighter but limited to roughly two kilovolts.

Because DEAs are capacitive rather than resistive, average current draw is modest — the design driver is the charge-discharge cycling loss, proportional to capacitance times voltage squared times frequency. Energy-recovery topologies that reclaim capacitive energy on discharge substantially improve the budget and are underused.

> **SAFETY**
>
> Kilovolt operation is not benchtop-casual. Use current-limited supplies with series resistors on every output, fully encapsulate electrodes in insulating elastomer, fit bleed resistors across every capacitor, interlock the enclosure, and observe the single-hand rule. Never operate an electroactive robot in water without verified encapsulation.

10.3 Cost of transport

Biological snakes achieve net costs of transport around 8 $J \cdot kg^{-1} \cdot m^{-1}$ sidewinding, 23 slithering, and 170 in concertina — the last figure explaining why snakes avoid concertina wherever possible. Robots are generally less efficient, and small soft robots considerably so. Consistent dimensionless cost-of-transport figures for soft snake robots are scarce in the published literature, which is a genuine reporting gap: papers report speed far more often than efficiency.

11 Performance Benchmarks

Metric	Soft snakes	Context
Speed	~1 body length/s (best)	Rigid snakes exceed this; most soft snakes are far slower
Actuation bandwidth	1–5 Hz pneumatic; 10–100 Hz electroactive	Pneumatic is valve and flow limited
Untethered duration	Minutes, where achieved at all	Rigid snakes achieve hours
Payload	Very low	Soft bodies cannot carry meaningful mass without sagging
Friction anisotropy achieved	$\eta \approx 1.5–3$	Biological baseline is 2 lateral, 4 backward
Model accuracy, quasi-static	~5% of body length	Achievable with well-identified Cosserat models
Model accuracy, locomotion speed	15–25%	Friction uncertainty dominates; 20% is respectable

Reporting conventions worth adopting: normalise speed by body length per second, normalise shape error by body length, and report cost of transport dimensionlessly. Report the regime where the model fails as well as where it succeeds — a paper stating that predictions hold to 15% below 3 Hz and degrade to 40% above it is more credible and more useful than one claiming uniform accuracy.

12 Applications

Soft snakes have not yet been fielded. The applications below represent the motivating cases; the parenthetical assessments reflect current readiness honestly.

- **Search and rescue.** Entering rubble voids too small and unstable for rigid machines. The motivating application for much of the field. *(Blocked by tether and autonomy duration.)*
- **Confined-space industrial inspection.** Pipes, ducts, turbine interiors, nuclear decommissioning. *(Rigid snakes already serve this; softness must justify itself against an incumbent.)*
- **Medical and surgical.** The most credible near-term soft application — endoluminal navigation, colonoscopy, magnetically steered catheters. Intrinsic safety matters most where the environment is living tissue, and short duration and external power are acceptable. *(Active commercial development.)*
- **Subsea inspection.** Water solves the power tether problem poorly but solves the gravity problem well, since neutral buoyancy removes the sag that limits soft snakes in air. *(Underexplored and arguably promising.)*
- **Planetary and extreme environments.** Lava tubes, ice fissures, glacial crevasses. *(Currently the domain of large rigid systems.)*
- **Agriculture.** Navigating foliage without damage. *(Early stage.)*

13 Open Problems

Untethered power

The dominant constraint. Pneumatic snakes have no adequate onboard compressor; electroactive snakes need kilovolt supplies that are only now becoming small enough. Any advance here unblocks everything downstream, and the recent progress in low-voltage dielectric elastomer chemistry and miniature high-gain converters makes this the field's most rapidly moving front.

Modelling fidelity against computational cost

Cosserat and finite-element models are accurate but too slow for reinforcement learning or model-predictive control at useful rates. Reduced models are fast but degrade under the non-uniform loading that ground contact always produces. Bridging this — through better reduced-order models, learned surrogates, or GPU-parallel continuum solvers — is a persistent and productive research direction.

Actuator bandwidth and force density

No current actuator is simultaneously fast, strong, soft and efficient. Pneumatics are strong but slow and tethered; dielectric elastomers are fast but weak and high-voltage; shape memory alloys are strong but thermally capped and inefficient. The gap between artificial and biological muscle remains wide.

Proprioception on a deforming body

Reliable, drift-free, distributed shape sensing that survives large strain and repeated deformation is unsolved. Self-sensing actuators are the most elegant partial answer.

Control of underactuated soft bodies

Far fewer actuators than deformation modes, hysteretic response, and unmodelled contact combine to defeat classical control. Learning-based methods work but transfer poorly and offer no guarantees.

Out-of-plane gaits and variable stiffness

Sidewinding, helical rolling and the dynamic weight redistribution that real snakes use all require holding shape against gravity while partially unsupported. Soft bodies sag. Variable-stiffness mechanisms — including antagonistic co-contraction, where two opposed actuators are energised together to raise effective stiffness without net curvature — are the most promising route and remain notably underexplored on snake platforms specifically.

Anisotropic skin engineering

Given that friction anisotropy is the propulsion mechanism, the sparseness of systematic work on soft anisotropic skins is striking. Achieving and maintaining ratios approaching the biological baseline on a compliant substrate, across varied terrain, is a well-posed and tractable problem with immediate impact on every soft snake built.

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Soft Snake Robots: Mechanical Design and Geometric Gait Implementation

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***Abstract*— Soft robots are compliant and durable, and thus potentially better-suited for swimming or crawling over rough terrain than conventional rigid robots. Biomimetic locomotion strategies are often implemented on soft robots, but are difficult to tune because of the complexity of mapping nonlinear mechanical systems to specific gait patterns. We propose an approach to soft robot control in which the robot is designed to physically realize a shape space (mechanically predefined body poses) to follow a theoretically optimal gait. Specifically in this paper we present an entirely soft snake robot designed to implement the prerequisite shape space for slithering gaits. When coupled with a geometric mechanics model relating serpentine shape sequences to displacement, appropriate gait patterns can be selected analytically based on the system's Lie bracket. We developed a modular, two-link soft robot and used the geometric mechanics model to identify several cyclic gait patterns for producing forward displacement. The gait patterns were tested on the soft robot and observed with a motion capture system to measure displacement and monitor shape sequences. Relative performance of the gait patterns is consistent with the geometric "soap-bubble method" as a heuristic for gait optimization, which demonstrates the applicability of this approach to soft robot control and coordination.**

I. Introduction & Related Work

Recent developments in soft robotics have expanded the possibilities of building biomimetic systems that are flexible, dynamic, and more robust in complex environments than their rigid counterparts. Soft robots are often composed of elastomeric actuators that are capable of continuum bending. This property makes them suitable for tasks in unstructured environments in which they can leverage the locomotion strategies of fish, snakes, and other animals that are relatively unconstrained by rigid skeletons. These examples from nature are high degree of freedom (DOF) systems that are difficult to approximate with rigid parts. For instance, snake robots have traditionally been built from a series of high DOF rigid actuators to produce sufficient bending capabilities. Although these efforts have been successful in demonstrating effective locomotion methods inspired by the gaits of real snakes, their movement requires a complex control system and is ultimately constrained by their inelastic bodies. The contributions we present in this work are two-fold: the mechanical design of a soft robot for undulatory locomotion, and demonstrating the utility of geometric mechanics to identify appropriate gaits for a soft continuum robot.

Snakes use a variety of strategies to achieve forward locomotion depending upon the composition of their habitats.

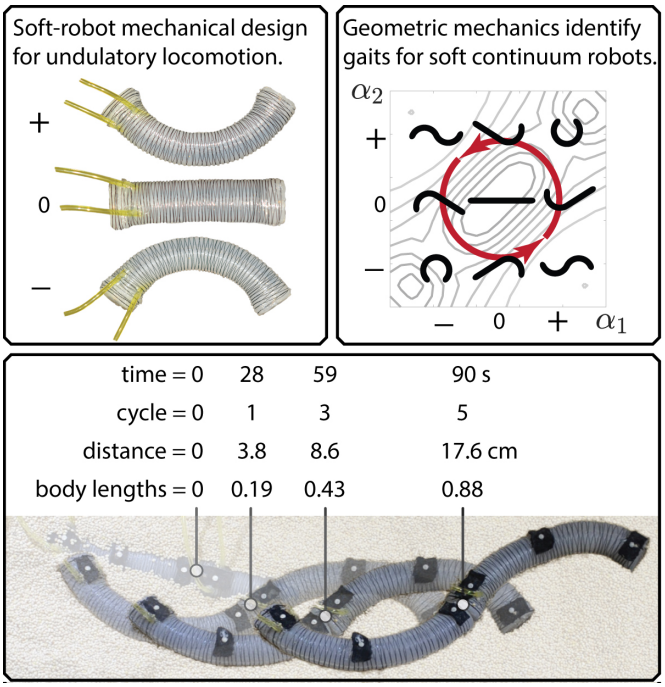


Fig. 1: The contributions of this work are two-fold: the mechanical design of soft-robotic actuators for undulatory locomotion, and the use of geometric mechanics to identify good candidate gaits for a soft continuum robot.

The most recognizable gait is *lateral undulation*, also known as serpentine locomotion or slithering [1]. This method requires a propagation of lateral bending along the snake's backbone, utilizing a resultant force vector to propel the snake forward upon surface contact. This gait does not itself result in forward movement without sufficient friction generated against the ground surface to produce lateral resistance, which is usually provided by anisotropic scales on the snake's underside. The need for scales is obviated when the ground medium has fluidic properties that produce a relatively high and anisotropic drag coefficient, such as water or sand [1].

Soft robots are difficult to control due to their intrinsic nonlinear behavior, and developing control strategies for soft robots is still a nascent area of research. Two primary approaches to this challenge have emerged in recent works: developing comprehensive system dynamics models to in-

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corporate into a control algorithm, and empirically designing soft mechanical systems to execute specific gait patterns. The first approach includes strategies such as recursive estimation of dynamics [2], low frequency Finite Element Analysis in real time [3], and training neural networks to yield an inverse kinetics function [4]. Most of these efforts focus on discovering forward or inverse mappings between actuator input and shape, and do not address the more complex issue of identifying and controlling shape sequences to produce an effective gait. Alternatively, evolutionary/learning algorithms can be used either to provide efficient gait "actions" directly [5], or to search through the parameters of a predetermined gait profile [6].

Empirically pre-programming the locomotion strategy in mechanical design has proven to be successful for many soft robots, including early works demonstrating quadruped gaits in [7] and [8]. More recently, a few groups have implemented effective biomimetic peristalsis gaits inspired by earthworms [9] and caterpillars [10] by centering the control strategy around a mechanical design that produces a gait pattern with simple actuation sequences. Our work builds upon this approach by designing a soft system that can assume a set of poses in the serpentine *shape space* to implement tunable gait patterns available to snake-like robots.

Snake robots have long been of interest to the robotics community for their applications in field robotics and confined spaces, and thus locomotion strategies for serpenoid robots are well-studied. Many biological snake gaits can be described as sinusoidal waves of bending propagating along the backbone in the ground and vertical planes, which creates lateral undulation and shifting contact points with the ground [11], [12]. A primary goal of geometric mechanics locomotion research is to quantify displacement as a function of such changes in body shape for simple snake-like systems [13]. Continuum bending can be approximated by a *hyper-redundant* system of rigid parts [14], which is the prevailing approach to implementing a snake-like robot. The most successful snake robots to date have been based on a design consisting of many modules actuated by servos and connected in series, with a rotational offset between adjacent modules to produce sinusoidal shapes in the ground and vertical planes [15]–[17]. Generating control algorithms based on the inverse kinematics of a hyper-redundant rigid robot is not a trivial task [14], [18], with challenges such as composing wave functions from key poses and mapping these functions to actuators in the snake's body [19]. One recent work implemented a lateral undulation gait in a soft robot, although they mounted these actuators on wheels to generate anisotropic friction between the soft body and hard surface for forward motion [20].

This paper proposes driving the design and control of an entirely soft snake robot based on a shape space derived from a geometric mechanics model for generating serpentine gaits. We define a two-dimensional shape space that encompasses a subset of serpentine shapes and derive a soft robot design from this definition that can assume a set of poses on this space. We developed silicone, pneumatic actuators that

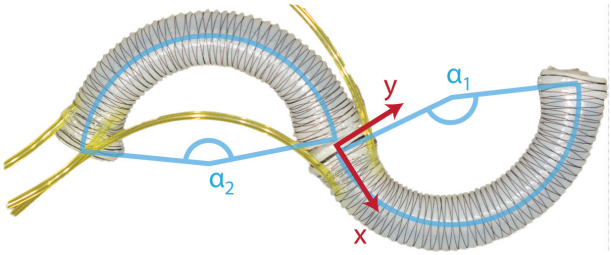


Fig. 2: Geometry of shape variables in two-actuator soft snake system. The origin of the body frame coordinate system is placed at the center point along the spine between the two actuators, with the x-dimension of position parallel to the spine and y-dimension perpendicular. The turning angle θ is defined relative to the x-axis.

are inexpensive, easy to manufacture, and capable of high-curvature continuum bending to realize pose sequences that compose a slithering gait using only a pair of antagonistic actuators. In addition, the actuators have a modular design and can be connected in series with embedded magnets, which offers the opportunity to produce more elegant backbone shapes for lateral undulation to be explored in the future.

Section II of this paper discusses the theory behind the geometric mechanics model used to inform the gait selection for the soft robot. Section III describes the actuator design choices and manufacturing process. The methodology for controlling and testing gaits is described in Section IV, and these results are compared with the outcomes predicted by the geometric mechanics model in Section V. Analysis of the project and plans to build upon these efforts are discussed in Section VI.

II. Snake Gait Selection

A primary goal of this work is to demonstrate the effectiveness of using a well-defined geometric gait model to develop a control sequence for a soft robot, and to show that this strategy can be separated from the nonlinear dynamics through which the system achieves the sequence of shapes. Our selection of shape sequences that are likely to form an effective gait is guided by the soap-bubble [21] model of gait performance which geometrically balances the displacement achieved over a gait cycle against the cost (measured in time or effort) required to execute that cycle.

The pose set for backbones of biological snakes can be represented by sinusoidal traveling wave functions [11], [19]. The *serpenoid-curve gait* for this system can be specified by the amplitude of sine and cosine curvature modes [21]. Our soft snake robot is a two-link, piecewise-continuous-curvature system which can approximate a subset of the biological snake's backbone shape space and thus can implement locomotion with select serpenoid-curve gaits. The soft robot's shape space can be parameterized by the arc angle of each link, α_1 and α_2. See Fig. 2 for an illustration of these shape parameters on the robot's body and Fig. 3(a) for example backbone shapes produced by varying α_1 and α_2. Displacement and orientation of the robot are measured as the displacement of the *body frame*, a coordinate system

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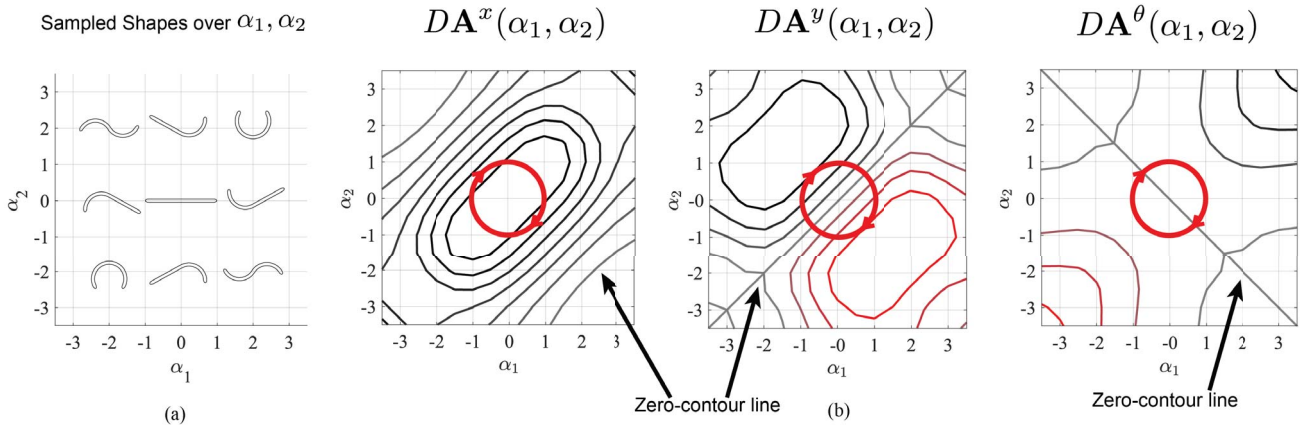


Fig. 3: (a) Sampled backbone shapes for the soft serpenoid system visualized as a function of α_1 and α_2. (b) Constraint curvature ($D\mathbf{A}(\alpha_1, \alpha_2)$) for the soft serpenoid system in a granular substrate with a lateral-to-longitudinal drag ratio of 1.8 to 1, visualized as contour plots with respect to x, y, and θ. An example gait stroke that varies α_1 and α_2 sinusoidally and 90 degrees out of phase to produce a "circular" stroke is plotted. Note that the gait cycle encloses part of a sign-definite region $D\mathbf{A}^x(\alpha_1, \alpha_2)$, indicating that there will be net displacement per cycle along the longitudinal axis of the body. The gait cycles encloses equal positive and negative regions on the contours $D\mathbf{A}^y(\alpha_1, \alpha_2)$ and $D\mathbf{A}^\theta(\alpha_1, \alpha_2)$ and thus will not produce any lateral or rotational displacement.

positioned at the center of the spine at $t = 0$ as shown in Fig. 2, from its initial position.

The configuration of the soft snake robot can be denoted by a position space G that describes the location and orientation of the robot in the world, and a shape space R that defines the relative placement of points on its body. The position of the robot is given by $g = (x, y, \theta)$ relative to a choice of origin frame, and the shape of the robot is given by $r = (\alpha_1, \alpha_2)$. We model the relationship between changes in shape and position as a *local connection* matrix $\mathbf{A}$,

$$\mathring{g} = -\mathbf{A}(r)\dot{r}$$

where $\mathring{g}$ is the system body velocity ($\dot{g}$ expressed in forward, lateral, and rotational components) and $\dot{r}$ is the rate of change of body shape. The model holds under the assumptions that gliding cannot occur and that displacement is locally proportional to the amount of shape change. A local connection matrix summarizing an appropriate dynamics model for this system can be constructed by taking a ratio between lateral and longitudinal drag forces and assuming quasistatic equilibrium [22]. The local connection matrix can be visualized as a set of vector fields on R relating changes in x, y, and θ to changes in shape $r = (\alpha_1, \alpha_2)$. Over a cyclic trajectory in the shape space (a gait), the net integral along the vector fields (i.e. the net displacement over the cycle) can be approximated by the area integral of the curvature of the constraints, $-D\mathbf{A}$, in the region enclosed by the gait. $D\mathbf{A}$ is the total Lie bracket of $\mathbf{A}$, and is closely related to its row-wise curl [23], [24].

The components of the constraint curvature $D\mathbf{A}(\alpha_1, \alpha_2)$ for the soft serpentine system are shown in Fig. 3 as contour plots, overlaid with an example gait that traverses the shape space by varying α_1 and α_2 sinusoidally and 90 degrees out of phase to produce a "circular" stroke. Note that there is an elliptical sign-definite region surrounding the origin in the contour plot for the x-component of position, whereas the y- and θ- plots have symmetric positive and negative regions around the origin. Gaits that enclose this region will thus produce net displacement along only the x-dimension of the body position. Previous work in geometric mechanics has demonstrated the utility of gait selection by choosing a stroke cycle that encloses sign-definite regions of these contour maps [17] [13] [22].

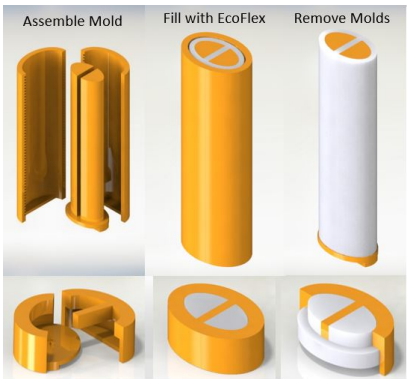
A gait cycle that encloses the entire sign-definite region should produce the largest displacement per cycle, but the shape sequence required to do this might have a high energy cost. The soap-bubble method [21] is an extension of this algorithm that attempts to optimize the gait cycle in terms of the displacement gained relative to the cost of the stroke. Curvature enclosed near the zero-contour is of low value, and so contributes relatively little to the net displacement and the shape changes required to encompass it require more time or effort in each cycle, meaning that a stroke that is contracted away from this contour is often more cost-efficient in terms of time and power.

We select gait cycles for our soft snake robot that enclose portions of the sign-definite region of the x-dimension contours of curvature to 1) demonstrate that the robot is able to achieve net longitudinal displacement with these gaits and 2) validate the displacement against the model based on the portion of sign-definite region enclosed by the cycle. The gaits tested all vary α_1 and α_2 in an elliptical stroke of the general form

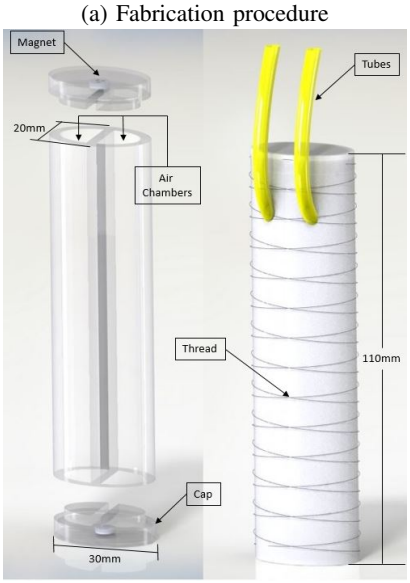
$$\begin{bmatrix}\alpha_1(t)\\ \alpha_2(t)\end{bmatrix} = R(\phi)\begin{bmatrix}a\cos(t)\\ b\sin(t)\end{bmatrix} \quad (1)$$

where $2a$ and $2b$ are the widths of the major and minor ellipse axes and $R(\phi)$ is a rotation of the ellipse about the origin by an angle of ϕ.

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(a) Fabrication procedure



(b) Assembly and dimensions

Fig. 4: (a) Fabrication procedure to mold the silicone actuators. (b) Depiction of actuator assembly along with the final wrapping configuration and tube placement.

III. Actuator Design & Characterization

To implement the gait sequences described above, the actuators must be capable of achieving a broad set of C- and S-shaped poses (see Fig. 3(a)). Each actuator must exhibit bidirectional bending to at least a 90° arc without producing significant twisting or rolling. The snake should be modular because locomotion will require at least one pair of antagonistic actuators in tandem, thus each actuator must have a connecting mechanism that does not require excessive use of rigid parts. Fiber-reinforced actuators were selected because they have been well characterized [25], and because they are highly modifiable. A typical fiber-reinforced actuator has one chamber used for actuation. A strain-limiting layer can be added for bending, and different fiber wrapping techniques can be used for extending or twisting [26].

In order to meet the requirements above, we implemented the following designs: 1) The mold structure was modified to create two air chambers for bidirectional bending. 2) A double-helical thread wrapping pattern was used to prevent twisting upon inflation, which disrupts the gait pattern significantly enough to prevent any forward displacement. 3) An elliptical cross section geometry was selected to approximate the typical body shape of a snake, and prevent the actuators from rolling during actuation. 4) Magnets were embedded into the silicone caps used to seal the ends of the chambers, so that an arbitrary number of actuators can be linked together. Placing them in the caps prevents them from interfering with inflation. The final actuator designed is 110mm in length, and has an elliptical cross section with a semi-major axis of 30mm, semi-minor axis of 20mm, and a wall thickness of 3mm.

The soft actuators were fabricated in 3D printed molds and wrapped and assembled by hand. The mold material was MeltInk3d 2.85mm PLA printed on an Ultimaker 2+ using a 0.6mm nozzle. The actuators were made from EcoFlex®00-30. As can be seen in Fig. 4, each actuator was molded in three parts: a main body containing the air chambers and two caps with embedded magnets. The caps were glued to both ends of the main body with SilPoxy® and the seam was sealed with a thin layer of the EcoFlex®00-30. The unique design of these fiber-reinforced actuators required several iterations of fabrication improvements. Once the final fabrication method was established, we characterized each actuator to ensure they had appropriate mechanical programming to realize the shape sequence involved in a lateral undulation gait.

During the iterative prototyping and testing procedure used to determine a final design for the soft actuators, it was confirmed that actuators that perform similarly in testing reproduced the desired gait pattern better. The final fabrication process greatly reduced the number of inconsistencies between actuators, but determining how each actuator curved under a specified pressure was necessary in implementing the gait pattern correctly. We used an Optitrack Prime 13 motion capture system to track the arc angle and length of each actuator at known values of input pressure, using a syringe to inflate each chamber from 0 to 24 kPA in 3.4 kPA increments. The results of these tests are summarized by the Pressure-Angle and Pressure-Length curves for the two closest performing actuators in Fig. 5.

By quantifying the relationship between pressure and curvature, we were able to select a pair of actuators with roughly symmetrical bending behavior out of the six actuators assembled with the final design. This characterization phase was essential to ensure that the robot would be capable of bending enough to attain the pose sequences of S- and C-shaped backbones, and also provided us with a mapping between input pressure and shape that helped with tuning the control algorithms to create shape sequences.

IV. Experimental Procedure

In order to implement gait strokes and evaluate their performance, we developed methodologies for directly controlling the shape parameters α_1 and α_2 of the robotic system, tracking progression through the shape space, and measuring net displacement of the robot. The experimental procedure we followed uses a motion capture system to

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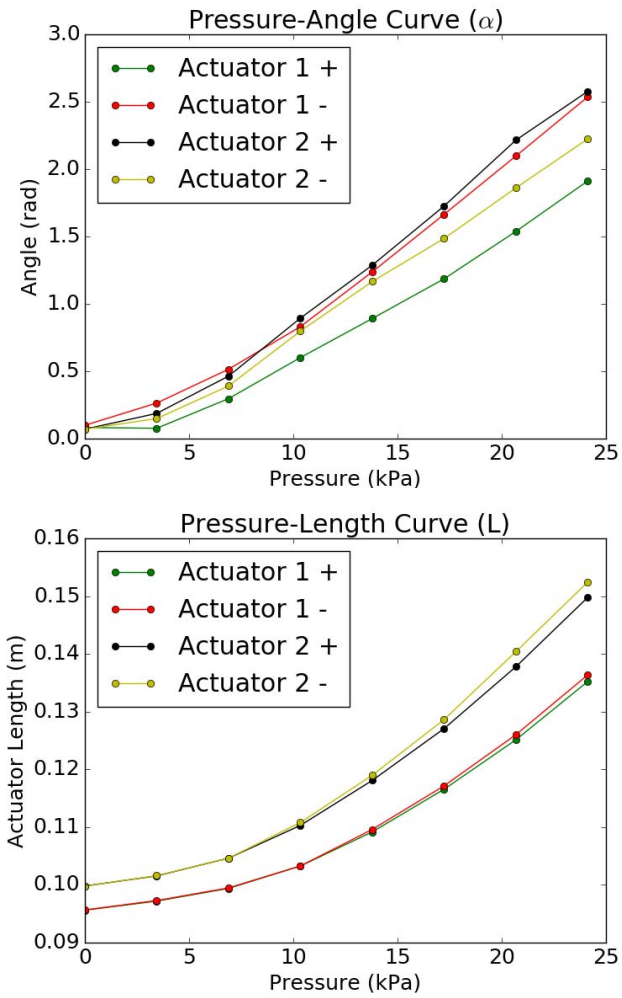


Fig. 5: Pressure-angle and pressure-elongation curves for each chamber of each actuator.

quantify the robot's shape and position changes for each gait stroke.

The snake's pneumatic control system is based on an open source design [27] that utilizes a single microcompressor with separate air lines for the actuator chambers. They can be driven to a separate pressure with solenoids valving at variable PWM rates. An Arduino Mega operates the solenoids by receiving control instructions from a separate computer via a serial data connection. Using the relationship between pressure and shape previously established, we developed Python scripts to drive PWM rates (and hence angles of curvature) sinusoidally in an appropriate pattern to realize elliptical gait stroke sequences.

We placed 4mm reflective markers in three locations on each actuator and employed the OptiTrack Prime system to monitor the bend angle and displacement of the snake. The markers were placed on the dorsal surface of each cap and at the center of the actuator's elastic spine (as seen in the shape snapshots in Fig. 6), so that the three points could be used to fit a circle to the actuator and thus yield a numerical approximation of its arc angle. The body frame of the two-actuator system was identified as the center point of the line formed by the two markers adjacent to each other where the actuators were clasped together by magnets. All motion capture data was saved and post-processed to compute and plot the body frame coordinates and curvature angles observed during a gait test.

A test area was created inside the motion capture stage by filling a large bin with millet, which was selected as the granular medium for these experiments because of its relatively low density ratio of 0.78 to 1 compared with the robot. The lateral-to-longitudinal drag ratio for the substrate was measured as 1.8 to 1. The drag ratio of the surface medium contributes considerably to net displacement, since lateral resistance translates lateral waves of bending along the backbone into forward movement, and so this ratio is crucial to estimating expected displacement.

We tested gait strokes described by Equation (1) with varying values of b and ϕ. The value of a was fixed in all tests to limit actuator bending to about 2 radians in either direction. Parameter values were selected empirically to enclose various regions within the zero-contour line, with rotations of $\phi = \frac{\pi}{4}$ or $\phi = \frac{3\pi}{4}$ to evaluate the impact of enclosing lengthwise and transverse regions of the elliptical area.

V. Results

Five different elliptical gaits were examined, with each test cycling through about 3 repetitions of a gait stroke during a total test duration of roughly 54 seconds or 18 seconds per stroke. Actual variations in shape parameters as observed with the motion capture system and calculated during post-processing of data are plotted on top of the $D\mathbf{A}^x(\alpha_1, \alpha_2)$ contour plot in Fig. 6 and Fi.g 7. The plot in Fig. 6 includes snapshots of backbone shapes that correspond to select points on the gait stroke to illustrate the pose sequence. Displacement is visualized as a function of time in Fig. 8. Net x- and y- displacement measurements for all gaits are shown in Table 9.

All gait strokes examined in these experiments achieved measurable forward displacement, and it should be noted that these gaits were within the boundaries of the sign-definite region with respect to $D\mathbf{A}^x$. For comparison with the displacement measurements observed with the motion capture system, we estimated the area integrals of $D\mathbf{A}^x$ based on the 1.8:1 drag ratio over each of the five regions enclosed by the actual shape trajectories. These predicted displacement values are also given in Table 9 in units of body lengths (L) per cycle. The cost for a gait stroke in this system was taken as the magnitude of the total shape change incurred over the cycle, which is the estimated perimeter length of the elliptical trajectory. To assess relative efficiency of each gait pattern, the average displacement was normalized against cost, and both cost and efficiency metrics are reported in Table 9 as well.

The circular stroke ($a : b = 1$) performed significantly better than the other gaits in terms of net displacement per cycle, and only somewhat better than the others in terms of efficiency. Net displacement appears to scale with size of the

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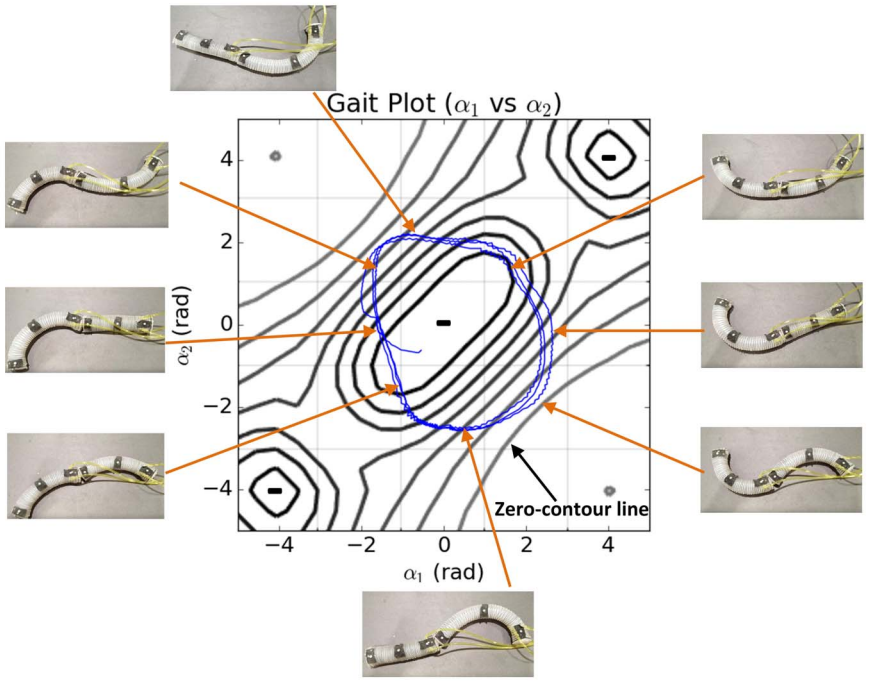
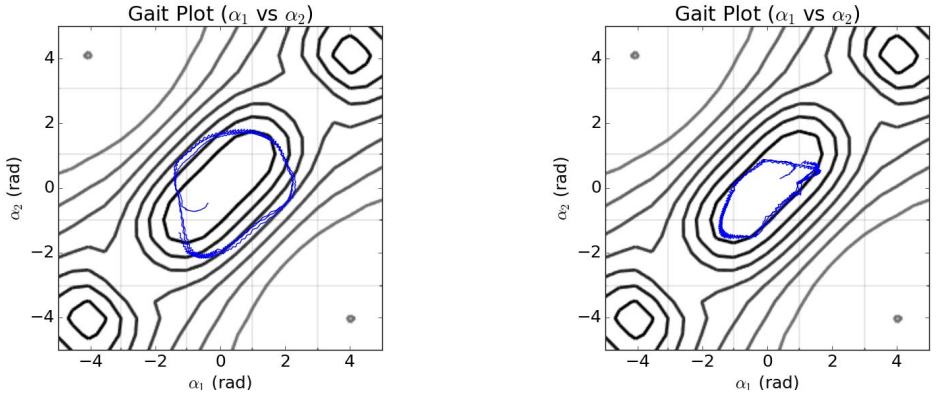


Fig. 6: Motion capture data from the soft robot demonstrating a circular gait stroke ($a : b = 1$) plotted on top of the $D\mathbf{A}_x(\alpha_1, \alpha_2)$ contour plot for the two-link soft serpenoid system. Key frames in the shape cycle are indicated with pictures.



(a) $a : b = 2, \phi = \frac{\pi}{4}$ (b) $a : b = 4, \phi = \frac{\pi}{4}$

(c) $a : b = 2, \phi = \frac{3\pi}{4}$ (d) $a : b = 4, \phi = \frac{3\pi}{4}$

Fig. 7: Gait stroke plots for four different elliptical gaits collected with the motion capture system. The observed gait strokes are plotted on top of the $D\mathbf{A}_x(\alpha_1, \alpha_2)$ contour plot.

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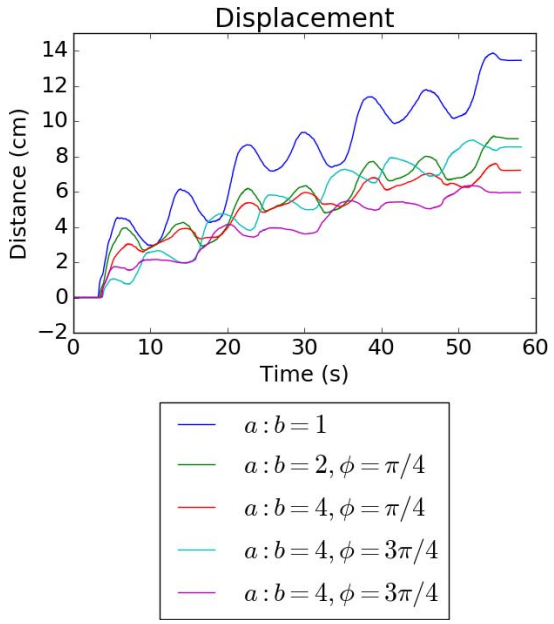


Fig. 8: Displacement along the longitudinal (x) body axis visualized as a function of time for all gait strokes examined with the motion capture system.

elliptical area enclosed about the origin and does not have a strong dependency on the orientation of that ellipse, although the elliptical gaits oriented at $\phi = \frac{\pi}{4}$ perform slightly better. It is worth noting that efficiency gains diminish as the size increases, which is consistent with the soap-bubble heuristic for gait selection. The measured displacements agree with the area integral predictions in terms of which gaits offer the most displacement, though not on the magnitude of the displacement. The soft snake actually produced more net forward movement by a factor of 2-3 over the predictions. This discrepancy is most likely attributed to a peristalsis effect caused by the cycles of elongation and retraction that accompany the bending in C- and S-shaped backbones. This explanation is also supported by the displacement plots and visual qualitative analysis of the gait, which show cycles of stretching and pulling forward on retraction. Another explanation could be that our drag coefficient measurement is incomplete which would effect the magnitude of the predicted displacement while having a minimal effect on the size of the sign-definite region.

A key trend to note across all gait plots is the contraction of the stroke curve in quadrants I and III. As shown in Fig. 6, a C-shaped backbone is produced when α_1 and α_2 have the same sign (quadrants I and III), and an S-shaped backbone when they have opposite signs (quadrants II and IV). The orientation of the elliptical sign-definite region of $D\mathbf{A}^x$ with its semi-major axis along $\alpha_1 = \alpha_2$ indicates that an "ideal" gait would include tight arc angles in the C-shapes and relatively shallow angles on the S-shapes. All gait plots gathered with motion capture show a somewhat concave curve in quadrant I, quadrant III, or both, highlighting that the soft robot was not as successful at achieving high-curvature C-shapes as it should be to perform a maximum-displacement gait stroke.

The strong S-shape backbones consistently demonstrate arc angles of about 2 radians from both actuators, which indicates that each is capable of achieving strong arcs in both positive-direction and negative direction curvature and suggests that the explanation for weaker C-shapes is unrelated to their bending capabilities. Forming a C-shape is usually a higher-cost operation than forming an S-shape of similar amplitude, since both ends of the body are moving in the same direction in a high-lateral-drag medium. We noticed the actuators pivoting slightly outwards about the magnetic fastener when forming C-shapes, probably at least partially due to this high drag force condition. Revising the clasp design to prevent this flattening of the shape could likely improve the C-shape pose by forcing the body to bend into the curve.

Additionally, the pair of actuators selected for the gait tests show some slight discrepancies in bending behavior and neither actuator is perfectly symmetrical, which could potentially be a source of error when implementing a gait pattern. A key challenge when working with soft actuators that are manufactured by molding is inconsistency in fabrication, which is difficult to avoid when using manual techniques. Fig. 5 shows enough variation in bend angle and elongation curves to potentially cause asymmetries in backbone curve.

VI. Conclusion & Future Work

We have successfully developed a modular soft snake robot that is capable of implementing gait cycles to mimic lateral undulation utilized by snakes for locomotion. By using a well-defined geometric gait model originally developed for rigid snake robots, we show that it is possible to select gaits that produce significant forward displacement separately from modeling the nonlinear dynamics of the soft actuators. The shape-space-driven design of the robot allowed for the successful realization of several parameterized gait strokes on the shape space. The performance of the gaits validated the soap-bubble method as a heuristic for selecting high-efficiency gaits, with those gaits that enclosed more of the sign-definite area yielding diminishing returns on efficiency even as the net displacement increased. The inherent properties of using soft actuators to mimic a soft-bodied animal proved advantageous for replicating serpentine locomotion, exceeding the net displacement estimates likely because of its natural tendency for peristalsis.

We plan to continue exploring shape-space-driven design with the addition of automated fabrication techniques such as 3D printing soft materials to enable production of more complex and consistent actuators. This would enable us to complete another design iteration of the actuators to improve the C-shaped backbone sequences to explore maximum-efficiency and maximum-displacement gaits. Additive manufacturing techniques could be used to investigate patterning the actuators to mimic the anisotropic friction that scales provide for biological snakes to produce interfacial traction on hard, flat surfaces. We plan to conduct similar experiments with longer chains of modular bidirectional continuum

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Gait Parameters	Measured Δx	Measured Δy	Average Disp (x)	Predicted Disp (x)	Cost	Efficiency
$a:b=1$	13.5 cm	3 cm	0.23 L/cycle	0.09 L/cycle	29.91 rad	0.0077 L/rad
$a:b=2,\phi=\frac{\pi}{4}$	9.0 cm	0.6 cm	0.15 L/cycle	0.07 L/cycle	22.96 rad	0.0065 L/rad
$a:b=4,\phi=\frac{\pi}{4}$	7.3 cm	0.4 cm	0.12 L/cycle	0.03 L/cycle	17.85 rad	0.0067 L/rad
$a:b=2,\phi=\frac{3\pi}{4}$	8.7 cm	1.6 cm	0.15 L/cycle	0.06 L/cycle	23.26 rad	0.0064 L/rad
$a:b=4,\phi=\frac{3\pi}{4}$	6.0 cm	0.25 cm	0.1 L/cycle	0.03 L/cycle	20.53 rad	0.0049 L/rad

Fig. 9: Net displacements along the longitudinal (x) and lateral (y) body axes during the gait tests. The displacements are measured over 3 cycles of the gait stroke and the time elapsed for each test was roughly 54 seconds. Average displacement in body lengths (L) per cycle are also given for comparison with displacements predicted by area integral of D**A**. The gait cost is the magnitude of shape change in a cycle (estimated perimeter of the actual elliptical trajectory, measured in radians), and efficiency normalizes average displacement against the cost of the stroke.

actuators to realize more elegant backbone shapes for other serpentine gaits. There are also plans to expand the geometric mechanics model to include backbone elongation so that it can better describe peristaltic systems. Finally, we will explore the effectiveness of evolutionary/learning algorithms to provide efficient gaits by both directly searching the gait space, and by searching the parameter space of a predetermined gait profile. Overall, this work validates using a soft actuator design to realize snake-like robot locomotion and geometric mechanics model as an approach to drive design and control of a soft robot.

Contributions

CB, CF, RH, and YM designed research. CB, CF, JR, and AK performed research. CB, CF, JR, and AK analyzed data. CB, CF, JR, KT, RH, and YM wrote the paper.

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